

Part II

Life Insurance

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7

Life Contingencies

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7.1 Introduction

This chapter aims to show how standard financial and actuarial mathematics calculations for life-contingent risks can be easily performed within the R statistical software and `lifecontingencies` package, Spedicato (2013a). While theoretical concepts will be briefly recalled during the exposition, the interested reader could find deeper details within standard actuarial references such as Finan (n.d.), Dickson et al. (2009), Bowers et al. (1997), and Ruckman & Francis (n.d.).

The chapter will be organized as follows: Section 7.2 will deal with financial mathematics applications, Section 7.3 will show life table analysis and applications, Section 7.4 will discuss pricing life insurance with R, while Section 7.5 will deal with life insurance reserves. More advanced topics will be discussed in Section 7.6. Finally, Section 7.8 will propose selected exercises for each previous section, whose solutions are reported in the demo files of the `lifecontingencies` package.

7.2 Financial Mathematics Review

Financial mathematics deals with the time value of the money. An amount X after t years will become $A(t) = X \cdot (1+i)^t$ if it follows the law of compound interest. i represents the effective rate of interest, that is, the amount of money that one unit invested at the beginning of a period will earn during that period, with interest being paid at the end of the period. Similarly, if the amount i had been earned at the beginning of the period, it would have been valued at $\frac{i}{1+i} = d$, d being the discount rate. A naive example follows: \$100 grows to

```
> 100 * ( 1 + 0.05 ) ^ 2
[1] 110.25
```

After 2 years, using a 5% interest rate, the discount rate would be

```
> 0.05 / ( 1 + 0.05 )
[1] 0.04761905
> interest2Discount( i = 0.05 )
[1] 0.04761905
```

which yields (for this 2-year contract)

```
> 110.25 * ( 1 - interest2Discount(i=0.05) ) ^ 2
[1] 100
```

Interest can be paid (reinvested) more frequently than once per period, being the fractions of a period called interest conversion periods. Often, the interest rate is expressed in nominal (also named “convertible”) terms, $i^{(m)}$, that is, the effective interest rate earned in the conversion period multiplied by m . A similar relationship exists for the discount rate; thus Equation (7.1) holds.

$$A(t) = (1+i)^t = \left(1 + \frac{i^{(m)}}{m}\right)^t = \left(1 - \frac{d^{(m)}}{m}\right)^{-t} = (1-d)^{-t} = v^{-t} \quad (7.1)$$

The `nominal2Real()` function computes the real interest rate given the nominal one:

```
> nominal2Real(i=0.06, k=12)
[1] 0.06167781
```

Rates of discount, for example, $d^{(m)} \rightarrow d$, can be given as input too, as the following code displays

```
> real2Nominal(i=0.04, k=12, type="discount")
[1] 0.04075264
```

Equation (7.1) can be used to perform basic financial mathematics calculations, for which functions within the `lifecontingencies` package help.

Example 7.1 Find the present value of \$1,000 to be paid at the end of 6 years at 6% per year payable in advance and convertible semiannually.

```
> 1000 * (1 - 0.06 / 2) ^ 12
[1] 693.8424
```

or equivalently,

```
> annualDiscount = nominal2Real(i=0.06, k=2, type="discount")
> i = discount2Interest(annualDiscount)
> presentValue(cashFlows=1000, timeIds=6, interestRates=i, probabilities=1)
[1] 693.8424
```

The Net Present Value (NPV) is a milestone concept within financial mathematics. It is the value in current money of a series of cash flows that are available at different points in time. It allows to value on a common basis different streams of cash flows. This means, in other words, ranking different investments according to their expected profit.

$$NPV = \sum_{j=1}^K CF_j (1 + i_j)^{-t_j} \quad (7.2)$$

Equation (7.2) shows its definition, while examples that follow display how the `lifecontingencies` package helps in calculating NPVs.

Example 7.2 Which investment is preferred between:

1. Invest \$500 now and receive \$100, 200, 300, 250 at the end of the following four years.
2. Invest \$700 now and receive \$1000 at the end of time 2.

```
> # example 2.1
> cfs1 = c(-500,100,200,300,250)
> times1 = 0:4
> NPV1 = presentValue(cashFlows=cfs1, t=times1, i=0.05)
> NPV1
[1] 241.4709
> # example 2.2
> cfs2 = c(-700,1000)
> times2 = c(0,2)
> NPV2 = presentValue(cashFlows=cfs2, t=times2, i=0.05)
> NPV2
[1] 207.0295
```

Thus, the first one is preferred here.

The inverse function of NPV is the Internal Rate of Return (IRR). The following example shows how to calculate IRR using R and the `lifecontingencies` package's functions.

Consider the following function whose minimum represents IRR:

```
> f<-function(irr) (presentValue(cashFlows=CF,
+   timeIds=TI,
+   interestRates=irr))^2

> CF = cfs1
> TI = times1
> IRR1 = nlm(f, 0.01)$estimate
> IRR1
[1] 0.2147816
```

This means that with a 21.478% rate of return, the discounted value of the cash flow is null.

```
> presentValue(cashFlows=cfs1, t=times1, i=IRR1)
[1] 0.0005410981
```

For the second investment, IRR is

```
> CF <- cfs2
> TI <- times2
> IRR2 <- nlm(f,0.01)$estimate
> IRR2
[1] 0.1952281
```

Financial mathematics developed formulas to price annuities that are a series of payments made at equal intervals of time. If payments are guaranteed to occur for a fixed period of time, the annuity is called “annuity-certain.” If payments are made at the end of each period for n periods, we have an “annuity-immediate or ordinary annuity. Otherwise, if payments are performed at the beginning of each period, we have an annuity-due. In addition, fractional payments can be performed within each period or for an infinite term (see Bowers et al. (1997) or Dickson et al. (2009) for further details).

As a remark, Equation (7.3) of an annuity-immediate of n -years term is

$$a_{\overline{n}|} = \frac{1 - (1+i)^{-n}}{i} \quad (7.3)$$

Example 7.3 *Price the following annuities:*

1. Calculate the present value of an annuity-immediate of amount \$100 paid annually for 5 years at a rate of interest of 9%.
2. Suppose a company issues a stock that pays a dividend at the end of each year of 10 indefinitely, and the company cost of capital is 6%. What would be the value of the stock at the beginning of the year?
3. What amount must you invest today at 6% interest rate compounded annually so that you can withdraw \$5,000 at the beginning of each year for the next 5 years?

For Example 3.1., the present value is

```
> i <- 0.09
> n <- 5
> 100*(1-(1+i)^(-n))/i
[1] 388.9651
```

which can be obtained using the `annuity` function,

```
> #example 3.1
> 100 * annuity(i=0.09, n=5, type="immediate")
[1] 388.9651
> #example 3.2
> 10 * annuity(i=0.06, n=Inf)
[1] 166.6667
> #example 3.3
> 5000 * annuity(i=0.06, n=5, type="due")
[1] 22325.53
```

Similarly, the accumulated value is computed by capitalizing the corresponding annuity. Equation (7.4) gives the accumulated value equation and symbol.

$$s_{\overline{n}|} = (1+i)^n a_{\overline{n}|} = \frac{(1+i)^n - 1}{i} \quad (7.4)$$

Example 7.4 *What amount will accumulate if we deposit \$5,000 at the beginning of each year for the next 5 years? Assume an interest of 6% compounded annually.*

From the relationship given above, one gets

```
> annuity(i=0.06, n=5, type="due")*5000*1.06^5
[1] 29876.59
```

or we can use the `accumulatedValue` function

```
> 5000 * accumulatedValue(i=0.06, n=5,type="due")
[1] 29876.59
```

Accumulated values are often used to perform financial planning for individuals, as the following example shows.

Example 7.5 *A man wants to save \$100,000 to pay for the education of his son in 10 years time. An education fund requires investors to deposit equal installments annually at the end of each year. If the guaranteed interest rate is 5%, how much does the man need to save each year in order to meet his target?*

```
> C <- 100000
> R <- C/accumulatedValue(i=0.05,n=10)
> R
[1] 7950.457
```

It is possible to define series of payments where amounts paid in each period are not constant. All such cash flows can be evaluated using the NPV Equation (7.2). However, closed formulas have been derived for some such sequences. The `lifecontingencies` package contains functions to evaluate the present value of arithmetically increasing, $(IA)_n$, or decreasing, $(DA)_n$, annuities, as shown by the following examples.

Example 7.6 *1. The following payments are to be received: \$500 at the end of the first year, \$520 at the end of the second year, \$540 at the end of the third years and so on, until the final payment is \$800. Using an annual effective interest rate of 2%, determine the present value of these payments at time 0.*

2. Determine the accumulated value of these payments at the time of the last payment.

In this example, we can either consider a fixed annuity of \$480, plus an arithmetical increase of \$20 (since the first payment of the increasing component will start with \$20, and not \$0),

```
> # example 6.1
> 480*annuity(i=0.02,n=16)+ 20*increasingAnnuity(i=0.02,n=16)
[1] 8711.431
```

For the accumulated value, one can use

```
> # example 6.2
> (480*annuity(i=0.02,n=16)+ 20*increasingAnnuity(i=0.02,n=16))*1.02^16
[1] 11958.93
```

or

```
> 480*accumulatedValue(i=0.02,n=16)+ 20*Isn(i=0.02,n=16)
[1] 11958.93
```

Similarly, the present value of annuities that grow following a geometric progression can be easily computed. If the payment grows by a rate of r and the interest valuation rate is i , the annuity can be calculated using a synthetic rate $j = \left(\frac{1+i}{1+r}\right) - 1$.

Example 7.7 *An annual annuity due pays \$1 at the beginning of the first year. Each subsequent payment is 5% greater than the preceding payment. The last payment is at the beginning of the tenth year. Calculate the present value using an annual effective interest rate of 4%.*

```
> j=(1+0.04)/(1+0.05)-1
> 1 * annuity(i=j,n=10,type="due")
[1] 10.44398
```

Annuities are said to be deferred if their inception date is different from the moment when they are being evaluated. The `lifecontingencies` package easily handles deferment.

Example 7.8 *Calculate the present value of an annuity due that pays annual amounts of \$1200 for 12 years with the first payment 2 years from now. The annual effective interest rate is 6%.*

```
> 1200*annuity(i=0.06,n=12,m=1,type="immediate")
[1] 9491.144
> 1200*annuity(i=0.06,n=12,m=2,type="due")
[1] 9491.144
```

The basic concepts presented up to now are used to perform financial calculations used in real business environments. One example is represented by bond pricing, which the following example shows.

Example 7.9 *A 5-year bond offers 3% coupons paid twice a year. The par value is \$100. Find its price using a 5% annual effective interest rate.*

A first strategy is to compute

```
> isemestral <- ( 1 + 0.05 )^0.5 - 1
> 100 * (1+0.05)^-5 + (100*0.03/2) * annuity(i=isemestral, n=10)
[1] 91.50142
```

but it is also possible to use

```
> 100 * (1+0.05)^-5 + 3 * annuity(i=0.05, n=5, k=2)
[1] 91.50142
```

A loan amortization program can be easily set up with `lifecontingencies` package help, as the following example shows.

Example 7.10 *A \$10,000 loan is stipulated with 7% APR and monthly installments for repayment. Find the monthly installment if the loan is planned to be repaid in 5 years.*


```

> R <- 10000/annuity(i=0.07,n=5,k=12)/12
> instalments <- rep(R,5*12)
> balance <- numeric(60)
> balance[1] <- 10000*(1+0.07)^(1/12)-R
> for(i in 2:60) balance[i] <- balance[i-1]*(1+0.07)^(1/12)-R
> amortization <- data.frame(month=1:60, rate=round(R,1),
+ balance <- round(balance,1))
> head(amortization)
  month rate balance
1     1  197  9859.5
2     2  197  9718.3
3     3  197  9576.2
4     4  197  9433.4
5     5  197  9289.7
6     6  197  9145.3
> tail(amortization)
  month rate balance
55    55  197   968.5
56    56  197   777.0
57    57  197   584.4
58    58  197   390.7
59    59  197   195.9
60    60  197    0.0

```

Finally, the `duration` and `convexity` functions can be used to calculate the duration and convexity of a series of cash flows. We will exemplify the use of such functions with a fictional 5-year term bond paying 3% coupon assuming a 5% yield rate. We will test Equation (7.5), D^* being the effective duration, $\Delta y = y_1 - y_0$ the variation in market yield, and P being the original price of the bond at initial yield y_0 .

$$\frac{\Delta P}{P} \approx -D^* \Delta y \quad (7.5)$$

```

> CFs = c(3,3,3,3,103)
> TIs = seq(1,5,1)
> bondPrice = presentValue(cashFlows = CFs,
+ timeIds = TIs, interestRates = 0.05)
> duration(cashFlows = CFs, timeIds=TIs, i=0.05,
+          macaulay=FALSE)
[1] 4.701744
> bondPrice2=presentValue(cashFlows = CFs,
+ timeIds = TIs, interestRates = 0.04)
> bondPrice2
[1] 95.54818
> bondPrice * (1 + duration(cashFlows = CFs,timeIds=TIs,
+ i=0.05,macaulay=FALSE) * 0.01)
[1] 95.63567

```

7.3 Working with Life Tables

Life insurance and annuities can be seen as the expected value of future cash flows currently valued, that is, actuarial present values (APV). In fact, we use the term *expectation* because APV is the expected value of a random variable that is itself a function of the future lifetime of the policyholder. While the present value of future cash flows is evaluated using the tools presented in Section 7.2, this section discusses how to handle life tables, from which the probabilities underlying actuarial mathematics calculations come.

Life tables consist of a nonincreasing sequence of l_x , $x = 0, 1, \dots, \omega$ that represents the number of subjects alive at the beginning of age x , ω being the terminal age. The number of subjects at time 0, l_0 , is the radix of the table.

Because a table counts survivors, one can compute a survival probability from l_x 's. If T_x denotes the (stochastic) remaining lifetime of some individual aged x , the probability that (x) reaches age $x + t$ is

$${}_tp_x = \mathbb{P}(T_x > t) = \frac{l_{x+t}}{l_x},$$

and the probability that (x) does not reach age $x + t$ is

$${}_tq_x = 1 - {}_tp_x = \mathbb{P}(T_x \leq t) = \frac{l_x - l_{x+t}}{l_x}.$$

As standard convention, ${}_1p_x$ will be denoted p_x .

Different ways can be approached to create a life table with the `lifecontingencies` package.

A direct approach is to impute directly x and l_x , as the code below shows:

```
> newLifeTable1 <- new("lifetable", x=seq(0,10,1),
+ lx=seq(from=1000,to=0,by=-100),name="Sample life table 1")
removing NA and 0s
```

or, one can provide 1-year survival/mortality rates (p_x , q_x) and build a life table using convenience functions as the code below shows:

```
> newLifeTable2 <- probs2lifetable(probs=seq(from=0.1,to=1,by=0.1),
+ radix=100000,type="qx",name="Sample life table 2")
```

S3 methods are available, like `head`, `tail`, `plot`, and `summary`.

```
> head(newLifeTable1)
  x  lx
1 0 1000
2 1  900
3 2  800
4 3  700
5 4  600
6 5  500
> summary(newLifeTable1)
This is lifetable:  Sample life table 1
Omega age is:  10
Expected curtated lifetime at birth is:  4.5
```

Moreover it is possible to export `lifetable` objects into `data.frame` objects as the code below shows:

```
> lifeTableDf <- as(newLifeTable1, "data.frame")
> class(lifeTableDf)
[1] "data.frame"
```

Demographic analysis can be performed using several functions within the package. The basic probabilities underlying life contingencies insurance are the probability that an insured aged x survives until age $x + t$, that is ${}_t p_x$, and its complement to one, ${}_t q_x$.

Example 7.11 *Using the Society of Actuaries illustrative life table, evaluate*

1. The probability that an insured aged 65 will die before reaching 85.
2. The probability that a policyholder aged 25 will survive until 65.

The Society of Actuaries (SoA) illustrative life table can be loaded using

```
> data(soa08Act)
```

Because it is a S4 class object (see Chapter 1), elements are obtained using the `@` symbol. For instance, the number of survivors aged 65 will be

```
> soa08Act@lx[soa08Act@x==65]
[1] 7533964
```

To compute the first probability, we can use either

```
> #example 11.1
> (soa08Act@lx[soa08Act@x==65]-soa08Act@lx[soa08Act@x==85])/
+ soa08Act@lx[soa08Act@x==65]
[1] 0.6869847
```

or, more conveniently, functions `pxt` and `qxt`

```
> qxt(soa08Act, 65,20)
[1] 0.6869847
> #example 11.2
> pxt(soa08Act, 25,40)
[1] 0.7876582
```

Similarly, the curtate expectation of life at age x , define as

$$e_x = \sum_{t=1}^{\infty} {}_t p_x \text{ and } e_{\overline{x:n}|} = \sum_{t=1}^n {}_t p_x$$

(expected life span between ages x and $x + n$) can be easily computed as shown in the code below.

Example 7.12 *Using SoA illustrative life table, calculate*

1. The curtate expectation of birth embedded in the life table.
2. The curtate expectation of life between 50 and 60 on the same life table.

For the (complete) curtate expectation of life, at birth, we just have to sum all the ${}_t p_0$'s, or equivalently, all the l_x 's except the first one, and divide it by l_0

```
> #example 12.1
> sum(soa08Act@lx[soa08Act@x%in%(1:110)]/soa08Act@lx[soa08Act@x==0])
[1] 71.34692
> sum(soa08Act@lx/soa08Act@lx[soa08Act@x==0])-1
[1] 71.34692
```

or use the `exn` function

```
> exn(object = soa08Act)
[1] 71.34692
```

For the expected life span between ages 50 and 60, we can obtain it also by summing some l_x 's:

```
> #example 12.2
> sum(soa08Act@lx[soa08Act@x%in%(51:60)]/soa08Act@lx[soa08Act@x==50])
[1] 9.583979
```

or using

```
> exn(object = soa08Act, x=50,n=60-50,type="curtate")
[1] 9.583979
```

The complete expectation of life between age x and $x + n$ can be expressed as

$${}^{\circ}e_x = \int_{t=0}^{\infty} {}_t p_x \text{ and } {}^{\circ}e_{x:n|} = \int_{t=0}^n {}_t p_x$$

or

$${}^{\circ}e_{x:n|} = \frac{{}_x L_x}{l_x}, \text{ where } {}_x L_x = \int_x^{x+n} l_s ds.$$

Example 7.13 Evaluate the complete expectation of life between ages 80 and 90.

The natural idea is to assume that when someone dies at age x , the death will occur at “age” $x + 1/2$. Thus, the complete expectation of life will be, here

```
> (sum(soa08Act@lx[soa08Act@x%in%(81:90)]/soa08Act@lx[soa08Act@x==80]))+
+ sum(soa08Act@lx[soa08Act@x%in%(80:89)]/soa08Act@lx[soa08Act@x==80]))/2
[1] 6.136299
```

or using the `exn` function

```
> exn(soa08Act,80,10,"complete")
[1] 6.136299
```

Another quantity relevant for demographers is the central mortality rate,

$${}_t m_x = \frac{{}_t d_x}{{}_t L_x} \text{ where } {}_t d_x = {}_t L_x - {}_{t+1} L_{x+1}.$$

Example 7.14 Evaluate ${}_2 m_8$ on the SoA illustrative life table.

```
> mxt(soa08Act,80,2)
[1] 0.0874816
```

`lifecontingencies` package allows the analyst to evaluate survival probabilities at fractional intervals, using different kinds of interpolation. For instance, with a linear interpolation, ${}_h p_x$ et ${}_{[h]+1} p_x$ (where $[h]$ is the integer part of $h \geq 0$),

$${}_h \tilde{p}_x = (1 - h + [h]) {}_{[h]} p_x + (h - [h]) {}_{[h]+1} p_x.$$

For a constant force of mortality, recall that

$${}_h p_x = \exp \left(- \int_0^h \mu_{x+s} ds \right), \text{ where } \mu_x \text{ is the force of mortality.}$$

Assume that $h \in [0, 1)$, and that $s \mapsto \mu_{x+s}$ is constant on $[0, 1)$; then

$${}_h \tilde{p}_x = \exp \left(- \int_0^h \mu_{x+s} ds \right) = \exp[-\mu_x \cdot h] = (p_x)^h.$$

Finally, if we still assume that $h \in [0, 1)$, another alternative is to assume that

$$\frac{1}{{}_h p_x} = \frac{1 - h + [h]}{{}_{[h]} p_x} + \frac{h - [h]}{{}_{[h]+1} p_x}.$$

or equivalently, we define

$${}_h \tilde{p}_x = \frac{{}_{[h]+1} p_x}{1 - (1 - h + [h]) {}_{[h]+1} q_x}.$$

Example 7.15 Evaluate the probability that a policyholder aged 80 1/4 will die within half a year, assuming:

1. Linear mortality interpolation.
2. Constant force of mortality behavior.

```
> #example 15.1
> pxt(object=soa08Act,x=80.25,t=0.5, fractional="linear")
[1] 0.959027
> #example 15.2
> pxt(object=soa08Act,x=80.25,t=0.5, fractional="constant force")
[1] 0.959027
```

In addition, the `lifecontingencies` package allows one to work with multiple lives. Consider two policyholders and their future lifetime random variables T_x, T_y . The joint life status is defined as $T_{xy} = \min(T_x, T_y)$, while the last survivor life status is defined as $T_{\overline{xy}} = \max(T_x, T_y)$. The joint survival probability with time horizon t is

$$\mathbb{P}(T_x > t \text{ and } T_y > t) = \mathbb{P}(T_{xy} > t) \text{ denoted } {}_t p_{xy}$$

while the last survivor probability with time horizon t is

$$\mathbb{P}(T_x > t \text{ or } T_y > t) = \mathbb{P}(T_{\overline{xy}} > t) \text{ denoted } {}_t p_{\overline{xy}}.$$

Assuming independent lifetime yields, ${}_t p_{xy} = {}_t p_x \cdot {}_t p_y$ and ${}_t p_{\overline{xy}} = {}_t p_x + {}_t p_y - {}_t p_{xy}$.

The example shown below displays how to perform demographic analysis with multiple heads.

Example 7.16 Assume the SoA illustrative life table applies to two policyholders aged 65 and 65, respectively. Find

1. The probability that the two policyholders will be both alive after 20 years.
2. The probability that at least one of them will be alive after 20 years.
3. The expected joint life period.

Consider here the same life table, for the husband and his wife (but different life tables can be considered).

```
> maleTable <- soa08Act; femaleTable <- soa08Act;
> tables <- list(male=maleTable,female=femaleTable)
```

For the joint probability, if we assume independent lifetimes, then the probability is

```
> #example 16.1
> pxt(tables$male,x=65,t=20)*pxt(tables$female,x=60,t=20)
[1] 0.1496391
```

but one can use the `pxyzt` function

```
> pxyzt(tablesList=tables, x=c(65,60),t=20,status="joint")
[1] 0.1496391
> #example 16.2
> pxyzt(tablesList=tables, x=c(65,60),t=20,status="last")
[1] 0.6414331
> #example 16.3
> exyzt(tablesList=tables, x=c(65,60),t=20,status="joint")
[1] 11.08887
```

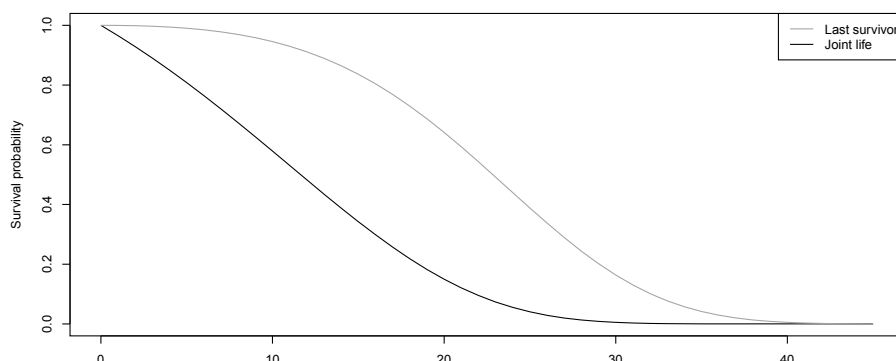
For these two individuals, it is possible to plot survival probabilities; see Figure 7.1.

```
> probjoint <- function(t){pxyzt(tablesList=tables, x=c(65,60),t,t,status="joint")}
> problast <- function(t){pxyzt(tablesList=tables, x=c(65,60),t,t,status="last")}
> vecT <- seq(0,45)
> plot(vecT,Vectorize(probjoint)(vecT),type="l",lty=2,ylab="Survival probability",xlab="")
> lines(vecT,Vectorize(problast)(vecT),col="grey",lwd=2)
```

7.4 Pricing Life Insurance

Life-contingent insurances are contracts that promise one or more payments upon occurrence of a life-contingent event. For example,

1. Life insurance contracts promise payments of a lump sum in case the insured dies within the period defined in the contract.
2. Annuity contracts pay a sum of money at the beginning (or the end) of each period, until the the term of the contract or the insured's death, whichever occurs first.
3. Endowment contracts pay a sum at the minimum between insured's death or the term of the contract.

**FIGURE 7.1**

Last survivor and joint life survival probability, when $x = 65$ and $y = 60$.

Pricing life insurance contracts is performed in several steps:

1. Define the financial and demographical assumptions, that is, the interest rate that will be used to discount future cash flows as well as the life table to estimate survival probabilities.
2. Determine the actuarial present value (APV) of life-contingent cash flows, that is, the actual price of the contractually promised benefits and eventual expense and profit loads.
3. Determine the premium(s), taking into account that it can be paid as a lump sum or within periodic installments.

Several examples will better explain such concepts. From now on, the following notation will be used in this chapter as well as the `lifecontingencies` package:

- x will represent insured's age at the inception of the contract.
- n will represent the term of the contract.
- m will represent the deferring period, that is, broadly speaking, the period since the benefits coverage would start (default value would be $m = 0$).
- i will represent the interest rate that discounts the contractually granted cash flows.
- k will represent the fractional payments frequency (default value would be $k = 1$).

All examples in this section will be based on the illustrative Society of Actuaries life table at an interest valuation rate of 6%. In addition, we will assume that benefits are paid at the end of each period. We remind interested readers of the cited references for deepening and additional examples.

The standard insurance contract that will be considered here is the n -year term life insurance policy that will pay \$1 at the death of the insured (x) if the death occurs before n years. The expected value of this contract is

$${}_nA_x = \sum_{k=1}^n \frac{\mathbb{P}(T_x \in [k-1, k))}{(1+i)^k} = \sum_{k=1}^n \frac{1}{(1+i)^k} \cdot {}_{k-1}p_x \cdot {}_1q_{x+k-1}. \quad (7.6)$$

Recall that the whole life contract is obtained when n is equal to ω , and the deferred contract when the sum is no longer from 1 to n , but from m to $n + m - 1$.

Before computer era, life insurance was evaluated using functions of the insureds' age tabulated in sheets called actuarial tables, the so-called "commutation functions." The `lifecontingencies` package offers functions to generate tables of commutation function.

Example 7.17 *From the SoA illustrative life table,*

1. *Generate the actuarial table object using an interest rate of 6%.*
2. *Export such an object in a dataframe showing corresponding commutation functions.*

```
> data(soaLt)
> #example 1
> soaAct <- new("actuarialtable", x=soaLt$x,
+ lx=soaLt$lx, interest=0.06)
> #example 2
> soaActDf <- as(soaAct, "data.frame")
> head(soaActDf)
```

	x	lx	Dx	Nx	Cx	Mx	Rx
1	0	10000000	10000000	168358017	47263.585	470300.9	12487975
2	1	9949901	9386699	158358017	44588.288	423037.4	12017674
3	2	9899801	8810788	148971318	42064.422	378449.1	11594637
4	3	9849702	8270000	140160530	39683.417	336384.6	11216188
5	4	9799602	7762203	131890531	37437.186	296701.2	10879803
6	5	9749503	7285396	124128328	6191.668	259264.0	10583102

The first example shows how to price a simple 3-year term life insurance policy. Assume the policyholder's age to be 36 and insured sum to be \$100,000. Using `lifecontingencies`' package functions, we can directly compute the APV of the selected insurance, $100,000 \cdot A_{36:\overline{3}|}^1$.

The probability to actually pay \$100,000 is, for each age,

```
> (probdeath <- -diff(soaActDf$lx)[soaActDf$x%in%36:38]/soaActDf$lx[soaActDf$x==36])
[1] 0.002140254 0.002274272 0.002420523
```

As the discount factors are, respectively,

```
> disc <- (1+0.06)^(-(1:3))
```

then the actuarial present value is

```
> sum(disc*probdeath) * 100000
[1] 607.5519
```

using Equation (7.6). Note that probabilities can also be obtained using (e.g., for the second one)

```
> k <- 2
> qxt(soaAct, x=36+k-1, t=1)*pxt(soaAct, x=36, t=k-1)
[1] 0.002274272
```

But all those computations can be done also using the `Axn` function:


```
> P <- 100000 * Axn(actuarialtable=soaAct, x=36, n=3)
> P
[1] 607.5519
```

Otherwise, using the “old” commutation functions, we would have written

$$A_{36:\overline{3}|}^1 = \frac{M_{36} - M_{36+3}}{D_{36}}.$$

```
> 100000 * with(soaActDf, (Mx[37]-Mx[40])/Dx[37])
[1] 607.5519
```

Another type of insurances consists of increasing or decreasing insurances with benefits payable at the end of the year of death. An n -year increasing term life insurance pays $k+1$ if the curtate future lifetime of the insured, $\tilde{K}_x$, is between $0, 1, \dots, n-1$. n -year term insurance, n -year increasing and decreasing term insurances are bounded by the relationship expressed in Equation (7.7).

$$(n+1)A_{x:\overline{n}|}^1 = (DA)_{x:\overline{n}|}^1 + (IA)_{x:\overline{n}|}^1 \quad (7.7)$$

```
> (10+1) * Axn(soaAct,60,10)
[1] 1.504674
> IAxn(soaAct,60,10) + DAxn(soaAct,60,10)
[1] 1.504674
```

Moreover, it can be assumed that the year can be divided into m subperiods and benefits to be paid on an m^{thly} basis, as the following examples show.

Example 7.18 Verify (numerically) $A_{x:\overline{n}|}^1{}^m = \frac{i}{i^m} A_{x:\overline{n}|}^1$.

```
> Axn(actuarialtable=soaAct,x=30,k=12)
[1] 0.1052721
> 0.06/real2Nominal(0.06,12)*Axn(actuarialtable=soaAct,x=30)
[1] 0.1052721
```

A whole life annuity-due is a series of payments made at the beginning of the year while the annuitant is alive. The whole life annuity for insured (x) has the following actuarial present value:

$$a_x = \sum_{k=1}^{\infty} \frac{\mathbb{P}(T_x > k)}{(1+i)^k} = \sum_{k=1}^{\infty} \frac{1}{(1+i)^k} \cdot {}_k p_x$$

The APV of \$100 payable at the beginning of each period to a 65-year-old policyholder until death would be

```
> sum((soaActDf$lx)[soaActDf$x%in%66:111]/soaActDf$lx[soaActDf$x==65]*
+ 1.06^(-(1:45)))
[1] 8.896928
```

which can be obtained using the function `axn`:

```
> axn(actuarialtable=soaAct,x=65)
[1] 9.896928
```

while if it were paid at the end of each period, it would be an annuity-immediate.

```
> axn(actuarialtable=soaAct,x=65, m=1)
[1] 8.896928
```

Another annuity example is the n -year temporary life annuity-due, where amounts of \$1 are paid at the end of each year as long as the policyholder survives, for up to a total of n years, or n payments otherwise.

Example 7.19 Find the APV of \$1000 paid by a policyholder between 25 and 45 years old at the beginning of each period.

```
> 1000 * axn(actuarialtable=soaAct, x=25, n=45-25)
[1] 12008.43
```

Sometimes annuities are deferred from the inception of the contract. In other words, the payment of the promised amount is deferred to the future and is contingent upon the survival of the subject.

Example 7.20 Find the APV of a whole life insurance on a policyholder aged 30, if it were deferred 10 years.

```
> Axn(actuarialtable=soaAct, x=30, m=10)
[1] 0.0882981
```

Handling geometrically increasing benefits is not complicated, as shown by the example that follows.

Example 7.21 A policyholder aged 50 wants to invest now a sum to purchase a deferred annuity-due with annual payment geometrically increasing by a ratio of 2% per annum. If the initial term of the annuity is 10,000, what is the amount the policyholder will be asked to pay, beyond any expense consideration?

Initially, we have to find the synthetic interest rate

```
> irate = 1.06 / 1.02 - 1
```

Then we can evaluate the annuity

```
> P=10000 * axn(actuarialtable=soaAct, x=50, i=irate)
> P
[1] 164275.2
```

Often, life annuities are paid m times a year. For example, a m -thly life annuity-due makes a payment of $\frac{1}{m}$ at the beginning of every m -thly period so that in 1 year the total payment is \$1.

Example 7.22 Find the APV of an annuity due if the annuitant's age is 60, assuming a fractional payment at the beginning of each month.

```
> axn(actuarialtable=soaAct, x=60, k=12)
[1] 10.68036
```

The `lifecontingencies` package easily handles fractional payments by determining the fractional payments cash flow pattern, their probabilities, and discounting them. Approximation formulas have been developed, for example, $\ddot{a}_x^{(m)} \approx \ddot{a}_x - \frac{m-1}{2m}$, but the algorithm within the package does not make use of them.

We have assumed, until now, the policyholder to want a policy with a single payment at the time the policy begins, being charged therefore the APV (also known as the net single premium) at the time of issuance. But, more frequently, the premium is paid in several installments, as following examples show.

Example 7.23 Find the benefit premium for a whole life insurance of a policyholder aged 25 paid at the beginning of the first 10 years the insured is alive, $_{10}P(A_{25}) = \frac{A_{25}}{\ddot{a}_{25:10}}$.

```
> P <- Axn(soa08Act,25)/axn(soa08Act,25,10)
> P
[1] 0.01052354
```

Modify the policy terms to take into account that deferring is easy.

Example 7.24 Calculate $P_{(20|\ddot{A}_{50})} = \frac{20|\ddot{A}_{50}}{\ddot{a}_{50}}$.

```
> P <- Axn(soa08Act,x=50,m=20)/axn(soa08Act,x=50)
> P
[1] 0.008945766
```

A more complex benefit premium computation is shown by the following example.

Example 7.25 For a special, fully discrete 5-year deferred whole life insurance of \$100,000 on a policyholder aged 40, it is assumed that the death benefit during the 5-year deferral period is the return of benefit premiums paid without interest and that the annual benefit premiums are payable only during the deferral period. Calculate the 5-year benefit premium. As a hint, the benefit premium, π , solves

$$\pi \ddot{a}_{40:\overline{5}|} = \pi \cdot (IA)_{40:\overline{5}|}^1 + 100000 \cdot {}_5|A_{40} \quad (7.8)$$

```
> pi <- 100000 * Axn(actuarialtable=soa08Act,x=40,m=5) /
+ (axn(actuarialtable=soa08Act,x=40,n=5)-
+ IAxn(actuarialtable=soa08Act,x=40,m=5))
```

We remind the interested reader of the cited literature to deepen the topic.

7.5 Reserving Life Insurances

The “benefit reserve” of an insurance policy consists of the amount of money the insurance company must have saved up to be able to provide for the future benefits of the policy, Finan (n.d.). Benefit reserves at time t are estimated using two methods, both assuming that the policyholder has survived at time t :

1. Under the prospective method, the benefit reserve is computed as the difference between the APV of future coverage benefits and the APV of future benefit premiums.

2. Under the retrospective method, the benefit reserve is computed as the difference between the accumulated value of past benefits paid less the accumulated value of past premiums received.

This chapter will only show examples regarding prospective benefit reserve calculation as well as fully discrete life-contingent policies. Equation (7.9) shows the general relationship that holds for benefit reserve. This is the so-called *prospective method*.

$$V_t = APV_t(\text{Future Benefits}) - APV_t(\text{Future Premiums}) \quad (7.9)$$

Examples will better exemplify such concepts. The benefit reserve for a whole life insurance is expressed by the following equation

$${}_tV_x = A_{x+t} - P \cdot \ddot{a}_{x+t}, \quad (7.10)$$

where P is the yearly benefit premium (fixed at time $t = 0$), $P = \frac{A_x}{\ddot{a}_x}$.

Example 7.26 Calculate the reserve for a (whole) life insurance policy on a 60-year-old policyholder paid by annual premiums at year 10.

First, we get the yearly benefit premium using

```
> P <- Axn(soa08Act,60)/axn(soa08Act,60)
```

and then we get the benefit reserve at year 10

```
> V <- Axn(soa08Act,60+10)-P*axn(soa08Act,60+10)
> V
[1] 0.2311365
```

The benefit reserve for an n -year temporary life insurance is expressed by the following equation

$${}_tV_x = {}_{n-k}A_{x+t} - P \cdot {}_{n-k}\ddot{a}_{x+t}, \quad (7.11)$$

where P is the yearly benefit premium (fixed at time $t = 0$), $P = \frac{{}_nA_x}{{}_n\ddot{a}_x}$.

Example 7.27 Calculate the reserve for a temporary life insurance policy on a 60-year-old policyholder paid by annual premiums for 30 years at year 10.

The yearly benefit premium will be

```
> P <- Axn(soa08Act,60,30)/axn(soa08Act,60,30)
```

and then, we get the benefit reserve at year 10

```
> V <- Axn(soa08Act,60+10,30-10)-P*axn(soa08Act,60+10,30-10)
> V
[1] 0.209061
```

If we want to visualize the evolution of the reserve, with time, the code will be

```
> V <- function(t) Axn(soa08Act,60+t,30-t)-P*axn(soa08Act,60+t,30-t)
> VecT <- seq(0,30)
> plot(VecT,Vectorize(V)(VecT),type="b")
```

Note there is also a *retrospective method* and an *iterative method* (the latter will be discussed at the end of this section). Because the APV of benefits and premiums are equal when the insured signs the contract (this is the principle of actuarial valuation), then

$$V_t < -APV_t(\text{Past Premiums}) - APV_t(\text{Past Benefits}). \quad (7.12)$$

This is the *retrospective method*.

In case the premium is paid for a limited number of periods, the following equation applies.

$${}_k^hV_{x:\overline{n}|}^1 = \begin{cases} A_{x+k} - {}_hP_{x:\overline{n}|}^1 * \ddot{a}_{x+h:\overline{n-h}|}, & k < h \\ A_{x+k}, & k \geq h \end{cases} \quad (7.13)$$

Example 7.28 Evaluate ${}_3^5V_{25:\overline{40}|}^1$ and ${}_{10}^5V_{25:\overline{40}|}^1$.

```
> #lump sum premium
> U <- Axn(soa08Act, x=25,n=40)
> #5-year benefit premium
> P <- U/axn(soa08Act, x=25,n=5)
> #Benefit reserves
> V3 <- Axn(soa08Act, x=25+3,n=40-3)-P*axn(soa08Act, x=25+3,n=5-3)
> ##at t=10
> V10 <- Axn(soa08Act, x=25+10,n=40-10) #benefit reserve at 10
> V3
[1] 0.03237489
> V10
[1] 0.06748181
```

Reserves for other life contingencies insurance follow a similar approach, that is, subtracting from the APV of prospective benefits the APV of prospective premiums.

Example 7.29 Calculate the reserve for a 20-year endowment at $t = 10$, for a policyholder aged 60.

```
> #Endowment insurance
> #full value premium
> U=AExn(soa08Act, 60,20)
> #benefit yearly premium
> P <- U/axn(soa08Act, 60,20)
> #reserve
> V <- AExn(soa08Act, 60+10,20-10)-P*axn(soa08Act, 60+10,20-10)
> V
[1] 0.355253
```

Example 7.30 Calculate the following reserves on annuities:

1. ${}_{10}V_{(20|\ddot{a}_{55})}$.
2. ${}_{30}V_{(20|\ddot{a}_{55})}$.

```
> #full value premium
> U <- axn(soa08Act,x=55,m=20)
> #yearly benefit premium
```

```

> P <- U/axn(soa08Act,x=55,n=20)
> #reserve 1
> V10 <- axn(soa08Act,x=55+10,m=20-10)-P*axn(soa08Act,x=55+10,n=20-10)
> V10
[1] 1.980102
> #reserve 2
> V30 <- axn(soa08Act,x=55+30)
> V30
[1] 4.698031

```

The idea of the *iterative method* is that

$$V_t = V_{t+1} - APV_t (\text{Future Premiums on}(t, t+1)) + APV_t (\text{Future Benefits on}(t, t+1)) \quad (7.14)$$

Thus, it is possible to use recursive techniques to compute V_t because at time $t = 0$, there should be no reserve. Giles (1993) suggested a simple algorithm to solve recursive equations.

Remark 7.1 *We do discuss here iterative equations for the reserves, but most actuarial quantities can actually be seen as solutions of such equations. For instance, A_x satisfies*

$$A_x = \frac{1}{1+i} (q_x + p_x \cdot A_{x+1}).$$

Similarly, $\ddot{a}_x$ is the solution of the following recursive equation,

$$\ddot{a}_x = 1 + \frac{1}{1+i} p_x \cdot \ddot{a}_{x+1},$$

and even e_x satisfies $e_x = p_x + p_x \cdot e_{x+1}$.

Consider a sequence $\mathbf{u} = (u_n)$ satisfying

$$u_n = a_n + b_n u_{n+1},$$

for all $n = 1, 2, \dots, m$ such that u_{m+1} is known, as well as both $\mathbf{a} = (a_n)$ and $\mathbf{b} = (b_n)$. The general solution is

$$u_n = \frac{u_{m+1} \prod_{i=0}^m b_i + \sum_{j=n}^m a_j \prod_{i=0}^{j-1} b_i}{\prod_{i=0}^{n-1} b_i} \quad (7.15)$$

with convention $b_0 = 1$. Then the following code can be used to solve this recursive equation:

```

> recurrent <- function(a,b,ufinal){
+ s <- rev(cumprod(c(1, b)))
+ return( (rev(cumsum(s[-1] * rev(a))) + s[1] * ufinal)/rev(s[-1]) )
+ }

```

Consider the temporary life insurance of Example 7.27. Then one can get that

$${}_tV_x = \frac{p_{x+t}}{1+i} \cdot {}_{t+1}V_x - P + {}_1A_{x+t}.$$

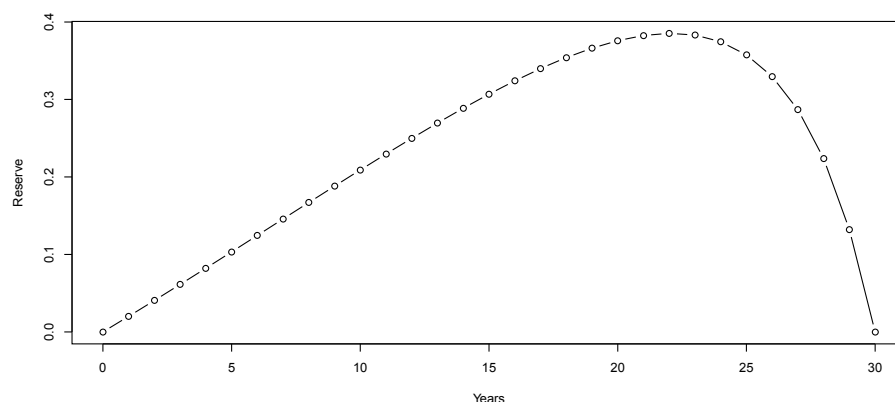
Then, based on notations of Equation (7.15), $u_{m+1} = 0$ with here $m = n$, and if the index of the recursive equation is t , then

$$a_t = {}_1A_{x+t} - P$$

and

$$b_t = \frac{p_{x+t}}{1+i}.$$

The code is then simply

**FIGURE 7.2**

Evolution of $t \mapsto {}_tV_x$ for a temporary life insurance over $n = 30$ years.

```
> P <- Axn(soa08Act,60,30)/axn(soa08Act,60,30)
> Vecta <- Vectorize(function(t) Axn(soa08Act,t,1))(60+0:29) - P
> Vectb <- Vectorize(function(t) pxt(soa08Act,t,1))(60+0:29)/1.06
> Vectv <- c(recurrent(a=Vecta,b=Vectb,ufinal=0),0)
> plot(0:30,Vectv,type="b")
```

And the graph is exactly the same as obtained on Figure 7.2.

7.6 More Advanced Topics

This section discusses, and exemplifies with R code, special and more advanced topics like contingent policies for multiple lives, consideration of expenses, and stochastic analysis.

The general theory about insurance and annuities can be applied to multiple lives. A key relationship regarding joint and last survival status is expressed in Equation (7.16), which holds also for APVs:

$$T_{xy} = T_x + T_y - T_{\overline{xy}}. \quad (7.16)$$

Example 7.31 Evaluate $A_{\overline{60:70}}$.

```
> #define the table
> coupleLifeTables=list(soa08Act,soa08Act)
> #direct approach
> Axyzn(tablesList=coupleLifeTables,x=c(60,70),status="joint")
[1] 0.5722832
> #indirect approach
> Axn(soa08Act,60)+Azn(soa08Act,70)-
+ Axyzn(tablesList=coupleLifeTables, x=c(60,70),
```

```
+ status="last")
[1] 0.5722827
```

Multiple life insurance theory is applicable in the evaluation of reversionary annuities, a special type of two-life annuities. A reversionary annuity pays benefits only when one of the lives has failed and then for as long as the other continues to survive. For example, the APV of an annuity that pays lifetime \$1 to y starting from the death of x is $a_{x|y} = a_y - a_{xy}$.

Example 7.32 *Compute the reversionary annuity on a 60-year-old policyholder, assuming the other insured is 70 years old.*

```
> revAnn <- axn(soa08Act,60)-axyzn(tablesList=coupleLifeTables,
+ x=c(60,70),status="joint")
> revAnn
[1] 3.589022
```

The second topic this section will discuss is the incorporation of expenses in calculating premiums and reserves. In practice, expenses are treated as if they were benefits. The gross premium G is determined as the sum of the APV of benefits and expenses. Similarly, the expense-augmented reserve is determined as the difference between the sum of prospective benefits and expenses and the APV of gross premiums.

Example 7.33 *For a fully discrete whole life insurance of \$100,000 on a policy holder aged 35 years, it is given that*

1. *Percent of premium expenses are 10% per year.*
2. *Per-policy expenses are \$25 per year.*
3. *Annual maintenance expense of \$2.50 for each \$1,000 of face value.*
4. *All expenses are paid at the beginning of the year.*

Find G . As a hint, Equation (7.17) shows the expression for G .

$$G\ddot{a}_{35} = 100,000A_{35} + \left(25 + \frac{2.5}{1000}100,000 + 0.1G\right)\ddot{a}_{35} \quad (7.17)$$

```
> G <- (100000*Axn(actuarialtable=soa08Act,x=35) +
+ (25 + 250)*axn(actuarialtable=soa08Act,x=35)) /
+ (0.9*axn(actuarialtable=soa08Act,x=35))
> G
[1] 1234.712
```

Example 7.34 *For a fully discrete whole life insurance of \$1,000 on a policyholder aged 45 years, it is given that:*

- *Expenses include 10% per year of gross premium.*
- *Additional expenses of \$3 per year.*
- *All expenses are paid at the beginning of the year.*

Calculate:

1. *The annual benefit premium.*

2. The annual gross premium.
3. The annual expense premium.
4. The benefit reserves at the end of year 1.
5. The total reserves at the end of year 1.

```
> #example 34.1
> U <- 1000*Axn(soa08Act,x=45); P=U/axn(soa08Act,x=45)
> P
[1] 14.25744
> #example 34.2
> G <- (U+3*axn(soa08Act,x=45))/((1-.1)*axn(soa08Act,x=45))
> G
[1] 19.17493
> #example 34.3
> E <- G-P
> E
[1] 4.917493
> #example 34.4
> V <- 1000*Axn(soa08Act,x=45+1)-P*axn(soa08Act,x=45+1)
> V
[1] 11.16089
> #example 34.5
> Vt <- 1000*Axn(soa08Act,x=45+1)+(0.1*G+3)*axn(soa08Act,x=45+1)
> -G*axn(soa08Act,x=45+1)
[1] -267.5782
> Vt
[1] 278.7391
```

The final topic covered in this section is the use of stochastic simulation of life-contingent insurances. It is worth remembering that life contingencies are themselves stochastic variables, as they are functions of the discount factor c and the future lifetime random variable. The `lifecontingencies` package contains functions that allow the analyst to generate variates from the expected future lifetime and life contingencies.

Example 7.35 Generate 10^3 variates of the curtate future lifetime at birth from the *SoA illustrative life table*.

Let us generate the variates using `rLife`:

```
> K0 <- rLife(n=10^3,object=soa08Act,x=0,type="Kx")
\begin{rcode}
From the law of large number, the average of those life times should not be
far away from the expected lifetime,\index{com}{t.test@\code{t.test}}
\begin{rcode}
> exn(soa08Act)
[1] 71.34692
> mean(K0)
[1] 73.129
> t.test(x=K0, mu=exn(soa08Act))
```

One Sample t-test

```

data: K0
t = 3.2432, df = 999, p-value = 0.001221
alternative hypothesis: true mean is not equal to 71.34692
95 percent confidence interval:
 72.05071 74.20729
sample estimates:
mean of x
 73.129

```

Generating variates from life contingencies can be used in pricing using the percentile premium principle approach. This approach prices the life contingency insurance finding the minimum premium that makes the insurer reduce the loss probability (the present value of prospective costs less the present value of prospective premiums) to not exceed a selected threshold.

Example 7.36 *A policyholder aged 25 purchases a 40-year term life insurance. Find the smallest premium the insurer could charge to set the probability of loss no greater than 5%.*

```

> samples <- rLifeContingencies(n=10^4, lifecontingency="Axn",
+ object=soa08Act, x=25, t=40, parallel=TRUE)

```

We can compute the classical benefit premium using

```

> APV <- Axn(soa08Act, x=25, n=40)

```

and percentile premium using

```

> percentilePremium <- quantile(x=samples, p=1-0.05)

```

We can also compare the two

```

> APV
[1] 0.0479709
> percentilePremium
 95%
0.2775051

```

Example 7.37 *A life insurance company is going to underwrite a pool of 1,000 retirees, all aged 65. Find the amount to be charged to each insured to make the insurance company not to lose with 99% probability.*

As a hint, remember that if $\tilde{Z}$ is the random variable representing the present value of the annuity, the present value of the total cost of the pool will be distributed as a normal distribution with parameters expressed by Equation (7.18)

$$\begin{aligned}\mu &= 1000 * \mathbb{E}(\tilde{Z}) \\ \sigma &= \sqrt{1000 * \text{var}(\tilde{Z})} .\end{aligned}\tag{7.18}$$

The Monte Carlo sample of A_{65} is obtained using

```

> ax65 <- rLifeContingencies(n=10^5, lifecontingency = "axn",
+ object = soa08Act, x=65, parallel=TRUE)

```

and the first two empirical moments are

```
> muax65 <- mean(ax65)
> sdax65 <- sd(ax65)
> qnorm(p=0.99,mean=1000*muax65,sd=(1000)^0.5*sdax65)/1000
[1] 10.16101
```

Finally, the `lifecontingencies` package can evaluate the APV of higher moments, by the use of the “power” parameter, as the following example shows.

Example 7.38 *Show that the variance of $(IA)_{45:\overline{20}|}^1$ as sample estimate is very close to its estimation by formula.*

```
> sampleIAxn <- rLifeContingencies(n=50000,lifecontingency="IAxn",object=soa08Act,
+ x=45, t=20,parallel=TRUE)
```

The sample variance is here

```
> var(sampleIAxn)
[1] 4.722363
```

which is comparable with the analytical one,

```
> IAxn(actuarialtable=soa08Act, x=45,n=20,power=2)-(IAxn(actuarialtable=soa08Act,
+ x=45,n=20,power=1))^2
[1] 4.719925
```

7.7 Health Insurance and Markov Chains

This section deals with the application of discrete Markov chains to Health Insurance. Discrete Markov chains represent a class of stochastic processes defined on a discrete set of possible states characterized by Equation (7.19). A classical application of Markov chains in insurance lies in Health and Disability coverage pricing and reserving. Another notable actuarial application of Markov chains is no-claim discount and bonus-malus policyholders’ evolution within a portfolio. Daniel (n.d.) provides an introduction to Markov chains application in the insurance business, while Haberman & Pitacco (1999) and Deshmukh (2012) provide a broader overview of actuarial topics, like health insurance, on which Markov chains can be effectively applied.

$$\mathbb{P}(X_{n+1} = x | X_1 = x_1, X_2 = x_2, \dots, X_n = x_n) = \Pr(X_{n+1} = x | X_n = x_n) \quad (7.19)$$

For a broader mathematical introduction, Ross (2009), Norris (1997), and Ching & Ng (2006) provide a good introduction to the topic.

7.7.1 Markov Chain with R

The R package `markovchain`, see (Spedicato 2013b), will be used throughout the chapter. The package vignettes provide much deeper detail on the package’s capabilities to handle and manipulate discrete Markov chains within an R environment, as well as how to perform probabilistic and statistical analyses on it.

```
> library(markovchain)
```

To show how to define a Markov chain within the package, let's consider a critical illness model with three states: healthy (H), critically ill (I), and dead (D). A Markov chain with associated transition probabilities can be defined as follows:

```
> stateNames <- c("H","I","D")
> cimMc <- new("markovchain",states=stateNames,
+ transitionMatrix=matrix(c(0.92,0.05,0.03,
+ 0.00,0.76,0.24,0.00,0.00,1.00),nrow=3, byrow=TRUE))
> cimMc
A Markov chain
A 3 - dimensional discrete Markov Chain with following states
H I D
The transition matrix (by rows) is defined as follows
      H      I      D
H 0.92 0.05 0.03
I 0.00 0.76 0.24
D 0.00 0.00 1.00
```

Various methods have been defined within the package to make probabilistic analysis easier, as the package vignettes reveal. For example, the probability that a subject initially healthy is dead in the fourth step is

```
> transitionProbability(cimMc^4, t0="H","D")
[1] 0.163991
```

Identifying steady probabilities vectors and absorbing states is easy as well:

```
> steadyStates(cimMc)
> absorbingStates(cimMc)
```

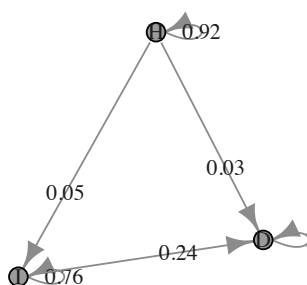
To perform such calculations, the power method, defined for the `markovchain S4` (see (Chambers & Hastie 1991) and Chapter 1) class has been used. Similarly, a `plotMc` method is available for the class, as Figure 7.3 displays, based on Csardi & Nepusz (2006).

```
> plotMc(cimMc)
```

If the transition probabilities between states vary by time, they can be modeled by non-homogeneous Markov chains, as the following example displays.

Example 7.39 *The status of residents in a Continuing Care Retirement Community (CCRC) is modeled by a nonhomogeneous Markov chain with three states: Independent Living (“H”), Health Center (“I”), and Gone (“D”). The transition probabilities are modeled in the following R code by $Q_0, Q_1, \dots, Q_3$.*

```
> Q0 <- new("markovchain", states=stateNames,
+ transitionMatrix=matrix(c(0.7, 0.2, 0.1,0.1, 0.6, 0.3,0, 0, 1),
+ byrow=TRUE, nrow=3))
> Q1 <- new("markovchain", states=stateNames,
+ transitionMatrix=matrix(c(0.5, 0.3, 0.2,0, 0.4, 0.6,0, 0, 1),
+ byrow=TRUE, nrow=3))
> Q2 <- new("markovchain", states=stateNames,
+ transitionMatrix=matrix(c(0.3, 0.2, 0.5,0, 0.2, 0.8,0, 0, 1),
```

**FIGURE 7.3**

Graph of the transition matrix `cimMc`.

```

+ byrow=TRUE,nrow=3))
> Q3 <- new("markovchain", states=stateNames, transitionMatrix=
+ matrix(c(0, 0, 1,0, 0, 1,0, 0, 1),byrow=TRUE, nrow=3))
> mcCCRC<-new("markovchainList",markovchains=list(Q0,Q1,Q2,Q3))

```

7.7.2 Valuation of Cash Flows

Two kinds of events result in payments: cash flow upon transitions, when payments are made upon transition from one state to another; and cash flows while in states represent payments made due to being in a certain state for a particular time period.

Example 7.40 Suppose a 2-year Accident and Death policy pays \$250 per year in case of illness and \$1000 in case of death. Suppose transition probabilities to be defined by `cimMc` matrix, the interest rate to be 5%. Suppose premiums are paid only when the policyholder is healthy, when no benefit is paid. Compute the APV.

Here, the subject is H

```

initialState<-c(1,0,0);attr(initialState, which="name")<-stateNames
v=1.03^-1

```

The possible transitions are the following:

```

> 250*transitionProbability(cimMc^2, "H","I")*v^2+
+ 1000*transitionProbability(cimMc^2, "H","D")*v^2+
+ 250*transitionProbability(cimMc, "H","I")*v+
+ 250*transitionProbability(cimMc, "H","I")*v+
+ 250*transitionProbability(cimMc, "I","I")*v^2+
+ 250*transitionProbability(cimMc, "H","I")*v+
+ 1000*transitionProbability(cimMc, "I","D")*v^2+
+ 1000*transitionProbability(cimMc, "H","D")*v
[1] 556.2494

```

Example 7.41 Each patient of the CCRC costs \$50 when healthy, \$200 when ill, \$10 when dead. Suppose the interest rate is 3%. Compute the APV of inpatient support using Monte Carlo simulation.

Consider the following function:

```
> getAPV<-function(mcList,t1="H"){
+   lifeStates<-character()
+   t2<-markovchainSequence(n=1,markovchain=mcList@markovchains[[1]],t0=t1) #state during second year
+   t3<-markovchainSequence(n=1,markovchain=mcList@markovchains[[2]],t0=t2) #state during third year
+   t4<-markovchainSequence(n=1,markovchain=mcList@markovchains[[3]],t0=t3) #state during fourth year
+   t5<-markovchainSequence(n=1,markovchain=mcList@markovchains[[4]],t0=t4) #state during fifth year
+   lifeStates<-c(t1,t2,t3,t4,t5)
+   APV<-0
+   v<-1.03^-1
+   for(i in 1:5){
+     value<-ifelse(lifeStates[i]=="H",50, ifelse(lifeStates[i]=="I",200,10))*v^((i-0.5))
+     APV<-APV+value
+     if(lifeStates[i]=="D") break}
+   return(APV)}

```

We can use that function , with 1,000 Monte Carlo simulations,

```
> simulations<-numeric(1000)
> set.seed(1)
> for(i in 1:1000) simulations[i]<-getAPV(mcCCRC)

```

and the average of those simulations is

```
> mean(simulations)
[1] 221.178

```

7.7.3 APV of Benefits and Reserves

Benefit premiums and reserves can be computed by applying the Equivalence Principle to any contingent payment situation, that is, equating the Actuarial Present Value of Premiums (APVP) to the Actuarial Present Value of the Benefits (APVB).

Example 7.42 Suppose a four-state homogeneous Markov model represents the joint mortality of a married couple: a husband and a wife. The states are 1 = husband alive, wife alive; 2 = husband dead, wife alive; 3 = husband alive, wife dead, and 4 = both husband and wife dead. A life insurer sells a 2-year term insurance contract to a married couple who are both age 60. The death benefit of \$100 is payable at the end of the year in which the second life dies, if both die within 2 years. Premiums are payable as long as at least one of them is alive and annually in advance. Interest rate $i = 5\%$. Calculate the annual benefit premium.

```
> mc2Lifes<-new("markovchain", states<-c("1","2","3","4"),transitionMatrix=matrix(
+   c(0.95, 0.02, 0.02, 0.01,
+     0.00, 0.90, 0.0, 0.10,
+     0.00, 0.00, 0.85, 0.15,
+     0.00, 0.00, 0.00, 1.00),byrow=TRUE,nrow=4))

```

The APVP and APVB are computed as follows, as premiums are paid at the beginning of period unless both subjects are dead.

```
> APVP<-1+(1-transitionProbability(mc2Lifes,"1","4"))*1.05^-1
> APVB<-100*(transitionProbability(mc2Lifes,"1","4")*1.05^-1
+   +(transitionProbability(mc2Lifes^2,"1","4")
-   -transitionProbability(mc2Lifes,"1","4"))*1.05^-2)
> P<-APVB/APVP
> P
[1] 1.167134

```

7.8 Exercises

The solutions to the following exercises are stored in demo files of the `lifecontingencies` package. For example, to load pricing section exercises, send the following code to R console:

```
> demo("pricing")
```

7.8.1 Financial Mathematics

- 7.1. Find the compound interest rate equivalent to a rate of compound interest of 8% payable semiannually.
- 7.2. In return for a payment of \$1,000 at the end of 10 years, a lender agrees to pay \$200 immediately, \$500 at the end of 6 years, and a final amount at the end of 15 years. Find the amount of the final payment at the end of 15 years if the nominal rate of interest is 5% converted semiannually.
- 7.3. Find the IRR of an investment that requires investing \$1,000 for 5 years and returning \$500 for the following 15 years.
- 7.4. What would you be willing to pay for an infinite stream of \$37 annual payments (cash inflows) beginning now if the interest rate is 8% per annum?
- 7.5. John receives \$400 at the end of the first year, \$350 at the end of the second year, \$300 at the end of the third year, and so on; the final payment is \$50. Using an annual effective rate of 3.5%, calculate the present value of these payments at time 0.
- 7.6. What is the difference between an annuity due with a 5-year term, interest rate 3% 12 times?
- 7.7. A bond whose term is 5 years, par value \$100, and yearly coupon rate is 5% prices at \$95.45. Find its yield.
- 7.8. Redo the duration example taking into account the convexity.

7.8.2 Demography

- 7.9. Create one lifetable for tables CL1 through CL3 shown in the `demoChina` dataset and compare the expected residual lifetime.
- 7.10. Find the probability of a life aged 2 to survive to age 4.
- 7.11. Find the number of people dying between 35 and 45.
- 7.12. Calculate the probability that three brothers aged 14, 15, 16 will all be alive after 60 years.

7.8.3 Pricing Life Insurance

Use `soa08Act` lifetable unless otherwise stated.

- 7.13. Create the Italian 92 male actuarial table at 3% from the `demoIta` dataset and name the table `SIM92`.

- 7.14. How much should a 25-year-old policyholder pay to receive \$100,000 when he turns 65, assuming he is alive?
- 7.15. How much should a 25-year-old policyholder pay to receive \$100,000 when he turns 65 or when he dies?
- 7.16. On the SOA actuarial table, calculate the APV whole life insurance for a policyholder aged 30 with benefit payable at the end of month of death at a 4% interest rate, $A_{30}^{(12)}$.
- 7.17. For the APV calculated in Exercise 7.16, determine the level benefit premium assuming it will be paid until the policyholder's death.
- 7.18. Calculate the quarterly premium that a policyholder aged 50 will pay until the year of death to insure a face value of \$100,000. The face value will be paid at the end of the quarter of death.
- 7.19. Compute the 15-year benefit premium for 35-year term insurance with benefit payable at the end of month of death. Assume the interest rate is 3% and the premium is to be paid at the beginning of each month.

7.8.4 Reserving Life Insurances

- 7.20. Calculate the reserve at time 4 for life insurance on a policyholder aged 60 paid with five yearly premiums as long as the policyholder is alive from year.
- 7.21. Calculate ${}_5V_{75:\overline{20}|}^{\frac{1}{}}$.
- 7.22. Calculate the benefit reserve for a 20-year endowment on a policyholder aged 60 at time 10 assuming level benefit premiums to be paid for 15 years.
- 7.23. Calculate the benefit reserve at time 10 for whole life insurance on 60 payable at the end of year of death with benefit premiums paid quarterly.

7.8.5 More Advanced Topics

- 7.24. Calculate a 10-year last survivor annuity-due on a policyholder aged 30 and on a policyholder aged 40.
- 7.25. Consider a 20-year term insurance on a policyholder aged 30, face value equal to \$1000. Calculate the gross level yearly premium, G , allowing for \$30 flat expenses and yearly maintenance expenses equal to 15% of G .
- 7.26. Sample a vector of size 10^4 from K_{25} and compute summary statistics.
- 7.27. Find the smallest premium to charge a policyholder aged 65 who wants to purchase an annuity-due in order to limit the probability of loss to 25%.

Prospective Life Tables

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8.1 Introduction

Prospective life tables depend on forecasting age-specific mortality. Considerable attention has been paid to methods for forecasting mortality in recent years. Much of this work has grown out of the seminal Lee–Carter method (Lee & Carter 1992). Other extrapolative approaches use Bayesian modelling, generalized linear modelling and state-space approaches. Methods for forecasting mortality have been extensively reviewed by Booth (2006) and Booth & Tickle (2008). This chapter covers various extrapolative methods for forecasting age-specific central death rates (also referred to as mortality rates). Also covered is the derivation of stochastic life expectancy forecasts based on mortality forecasts.

The main packages on CRAN for implementing life tables and mortality modelling are `demography` (see Hyndman 2012) and `MortalitySmooth` (see Camarda 2012), and we will concentrate on the methods implemented in those packages. However, mention is also made of other extrapolative approaches, and related R packages where these exist.

We will use, as a vehicle of illustration, U.S. mortality data. This can be extracted from the Human Mortality Database (2013) using the `demography` package.

```
> library(demography)
> usa <- hmd.mx("USA", "username", "password", "USA")
```

The `username` and `password` are for the Human Mortality Database.

The `hmd.mx` function downloads all available annual data by single years of age. Note that it is also possible to download data in 5-year age groups and that the methods described in this chapter can accommodate 5-year age groups. We use single years of age and annual years over time.

The object `usa` contains all the necessary data to forecast mortality:

```
> names(usa)
[1] "type" "label" "lambda" "year" "age" "pop" "rate"
```

A useful summary is given by

```
> usa
Mortality data for USA
  Series: female male total
  Years: 1933 - 2010
  Ages:  0 - 110
```

`usa` is a 'demogdata' object of type 'mortality'. The functions in the `demography` package are designed to use `demogdata` objects.

In order to stabilize the high variance associated with high age-specific rates, it is necessary to transform the raw data by taking logarithms. Consequently, the mortality models considered in this chapter are all in log scale.

8.2 Smoothing Mortality Data

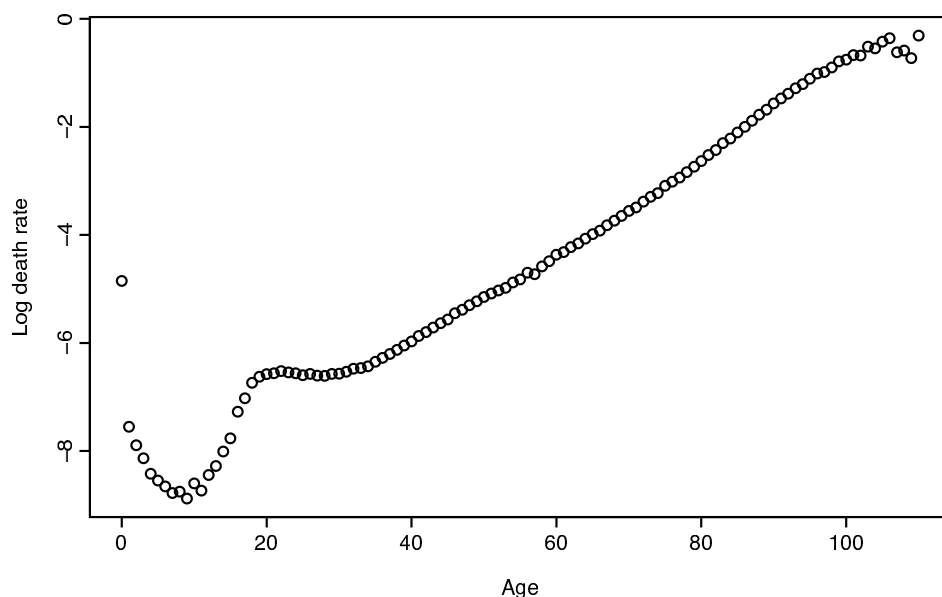
Suppose $D_{x,t}$ is the number of deaths in calendar year t of people aged x , and $E_{x,t}^c$ is the total years of life lived between ages x and $x + 1$ in calendar year t , which can be approximated by the mid-year (central) population at age x in year t .

Mid-year populations can be obtained from the `usa$pop` list:

```
> names(usa$pop)
[1] "female" "male" "total"
```

For example, the matrix `usa$pop$total` contains the total population at age x (per row) at mid-year t (per column):

```
> usa$pop$total[1:5,1:6]
      1933    1934    1935    1936    1937    1938
0 1963038 1920566 1940669 1953470 1967325 2012973
1 2034701 1927298 1896434 1924241 1945812 1965169
```

**FIGURE 8.1**

Male mortality rates for single years of age, United States 2003.

```
2 2178069 2106591 2037580 2020441 2063568 2102951
3 2229949 2191411 2126538 2056968 2040318 2086125
4 2274434 2226345 2187956 2122040 2053175 2038711
```

Similarly, the observed mortality rates, defined as

$$m_{x,t} = D_{x,t}/E_{x,t}^c,$$

are obtained from the list of matrices `usa$rate`. For example, rates for the total population are obtained by

```
> usa$rate$total[1:5,1:6]
      1933      1934      1935      1936      1937      1938
0 0.061667 0.067866 0.061964 0.062786 0.061008 0.058018
1 0.009459 0.010622 0.008831 0.008895 0.008366 0.007793
2 0.004351 0.004852 0.004171 0.004199 0.004000 0.003563
3 0.003104 0.003230 0.002980 0.002870 0.002681 0.002463
4 0.002386 0.002451 0.002404 0.002272 0.002151 0.001912
```

Figure 8.1 shows mortality rates for U.S. males in 2003, based on the following R code:

```
> plot(usa$age, log(usa$rate$male[, "2003"]), xlab= "Age", ylab="Log death rate")
```

Because `usa` is a 'demogdata' object, this graph can also be obtained by

```
> plot(usa, series="male", years=2003, type="p", pch=1)
```

This example shows that the mortality rates follow a smooth function with some observational error. The observational error has higher variance at very old ages (because the populations are small) and at childhood ages (because the mortality rates are low). (Note

that observed mortality rates at very old ages can exceed 1 because the number of deaths at age x may exceed the mid-year population aged x .)

Thus, we observe $\{x_i, m_{x_i, t}\}$, $t = 1, \dots, n$, $i = 1, \dots, p$, where

$$\log m_{x_i, t} = f_t(x_i^*) + \sigma_t(x_i^*)\varepsilon_{t, i},$$

$\log$ denotes the natural logarithm, $f_t(x)$ is a smooth function of x , x_i^* is the mid-point of age interval x_i , n is the number of years and p is the number of ages in the observed data set, $\varepsilon_{t, i}$ is an iid random variable and $\sigma_t(x)$ allows the amount of noise to vary with x .

Then the observational variance, $\sigma_t^2(x)$, can be estimated assuming deaths are Poisson distributed (Brillinger 1986). Thus, $m_{x, t}$ has approximate variance $D_{x, t}/(E_{x, t}^c)^2$, and the variance of $\log m_{x, t}$ (via a Taylor approximation) is

$$\sigma_t^2(x) \approx 1/D_{x, t}.$$

Life tables constructed from the smoothed $f_t(x)$ data have lower variance than tables constructed from the original $m_{t, x}$ data, and thus provide better estimates of life expectancy. To estimate f , we can use a nonparametric smoothing method such as kernel smoothing, loess, or splines. Two smoothing methods for estimating $f_t(x)$ have been widely used, and both involve regression splines. We will briefly describe them here.

8.2.1 Weighted Constrained Penalized Regression Splines

Hyndman & Ullah (2007) proposed using constrained and weighted penalized regression splines for estimating $f_t(x)$. The weighting takes care of the heterogeneity due to $\sigma_t(x)$, and a monotonic constraint for upper ages can lead to better estimates.

Following Hyndman & Booth (2008), we define weights proportional to the approximate inverse variances, $w_{x, t} \propto D_{x, t}$, and use weighted penalized regression splines (Wood 2003, He & Ng 1999) to estimate the curve $f_t(x)$ in each year. Weighted penalized regression splines are preferred because they can be computed quickly and allow monotonicity constraints to be imposed relatively easily.

We apply a qualitative constraint to obtain better estimates of $f_t(x)$, especially when $\sigma_t(x)$ is large. We assume that $f_t(x)$ is monotonically increasing for $x > b$ for some b (say, $b = 65$ years). This monotonicity constraint allows us to avoid some of the noise in the estimated curves for high ages, and is not unreasonable for this application (after middle age, the older you are, the more likely you are to die). We use a modified version of the approach described in Wood (1994) to implement the monotonicity constraint.

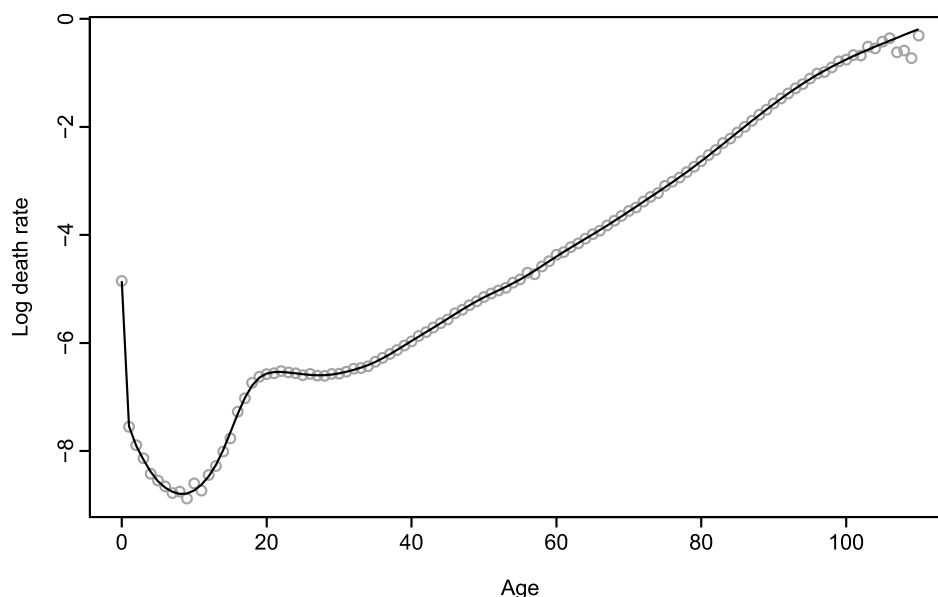
Figure 8.2 shows the estimated smooth curve, $f_t(x)$, for the U.S. male mortality data plotted in Figure 8.1. This is easily implemented in the `demography` package using the following code:

```
> smus <- smooth.demogdata(usa)
> plot(usa, years=2003, series="male", type="p", pch=1, col="gray")
> lines(smus, years=2003, series="male")
```

8.2.2 Two-Dimensional P-Splines

The above approach assumes $f_t(x)$ is a smooth function of x , but not of t . Hyndman & Ullah (2007) argued that the occurrence of wars and epidemics meant that $f_t(x)$ should not be assumed to be smooth over time. In the absence of wars and epidemics, it is reasonable to assume smoothness in both the time and age dimensions. For this reason, we use data from 1950 in this subsection.

```
> usa1950 <- extract.years(usa, years=1950:2010)
```

**FIGURE 8.2**

Smoothed male mortality rates for single years of age, United States 2003. The smooth curve, $f_t(x)$, is estimated using weighted penalized regression splines with a monotonicity constraint for ages greater than 65.

Currie et al. (2004) proposed using two-dimensional splines. We will call this approach the Currie-Durban-Eilers or CDE method.

The CDE method adopts the generalized linear modelling (GLM) framework for the Poisson deaths $D_{x,t}$ with two-dimensional P-splines. Here

$$D_{x,t} \sim \mathcal{P}(E_{x,t}^c \cdot \exp[s(x,t)])$$

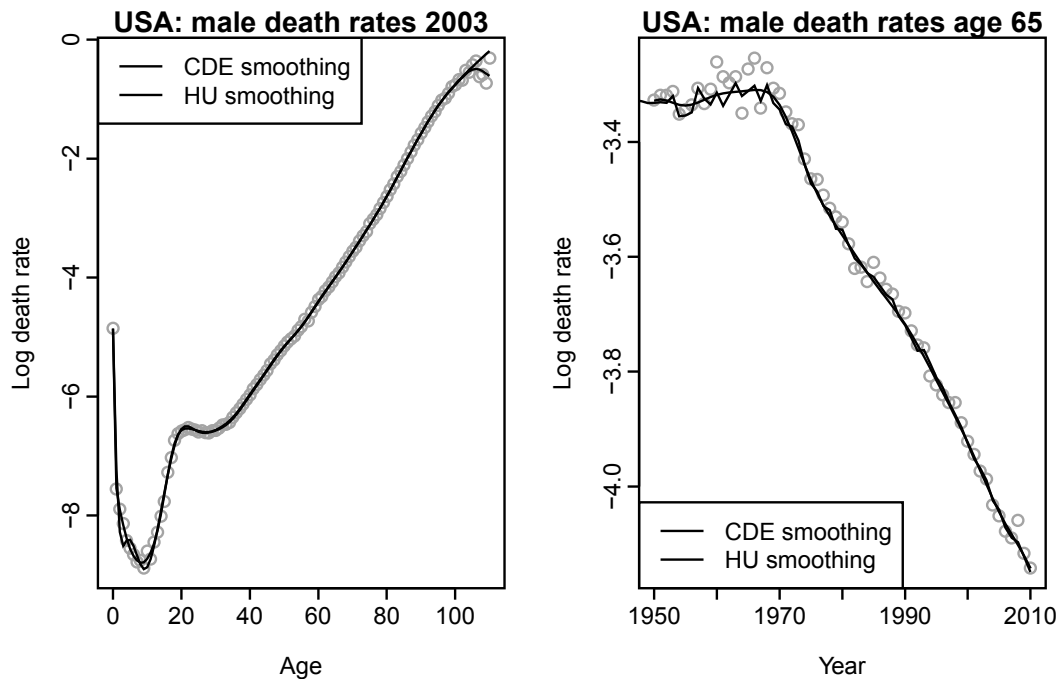
where the number of deaths, $D_{x,t}$, will be proportional to the exposure, $E_{x,t}^c$, and $f_t(x) = \exp[s(x,t)]$ will denote the mortality rate.

This is implemented in the `MortalitySmooth` package in R (Camarda 2012) and compared with the Hyndman & Ullah (2007) approach using the following code:

```
> library(MortalitySmooth)
> Ext <- usa1950$pop$male
> Dxt <- usa1950$rate$male * Ext
> fitBIC <- Mort2Dsmooth(x=usa1950$age, y=usa1950$year, Z=Dxt, offset=log(Ext))

> par(mfrow=c(1,2))
> plot(fitBIC$x, log(usa1950$rate$male[, "2003"]), xlab="Age", ylab="Log death rate",
+ main="USA: male death rates 2003", col="gray")
> lines(fitBIC$x, log(fitBIC$fitted.values[, "2003"])/Ext[, "2003"])
> lines(smus, year=2003, series="male", lty=2)
> legend("topleft", lty=1:2, legend=c("CDE smoothing", "HU smoothing"))

> plot(fitBIC$y, log(Dxt["65", ]/Ext["65", ]), xlab="Year", ylab="Log death rate",
+ main="USA: male death rates age 65", col="gray")
> lines(fitBIC$y, log(fitBIC$fitted.values["65", ]/Ext["65", ]))
```

**FIGURE 8.3**

Smoothed male mortality rates using bivariate P-splines, United States.

```
> lines(smus$year, log(smus$rate$male["65",]), lty=2)
> legend("bottomleft", lty=1:2, legend=c("CDE smoothing", "HU smoothing"))
```

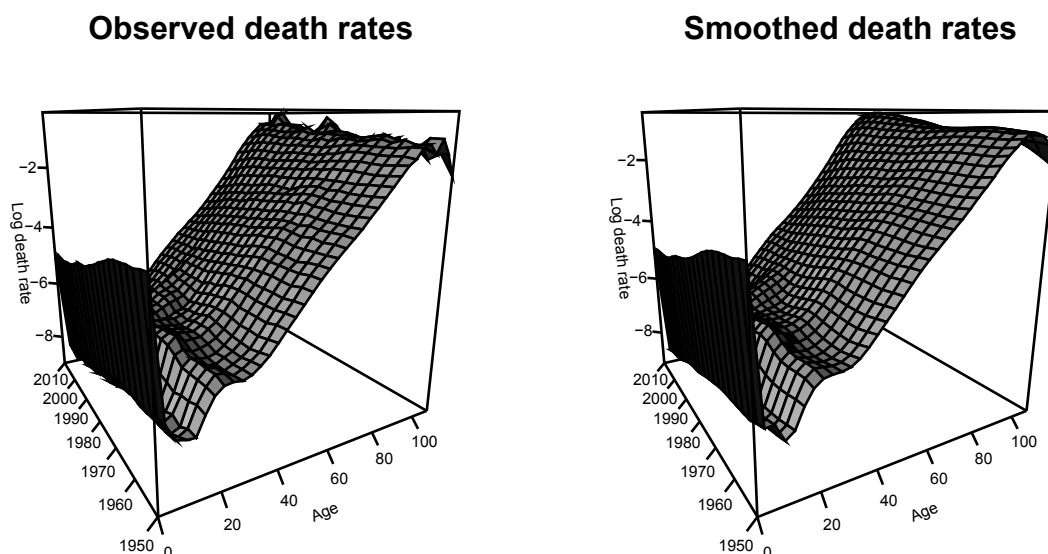
Figure 8.3 shows the estimated smooth curve, $f_t(x)$, for the U.S. male mortality data using the bivariate P-spline (CDE) method (Currie et al. 2004) and the univariate penalized regression spline method of Hyndman & Ullah (2007). Note that the univariate method is not smooth in the time dimension (right panel), but gives a better estimate for the oldest ages due to the monotonic constraint.

```
> par(mfrow=c(1,2))
> year=usa1950$year[seq(1,62,by=3)]
> age=usa1950$age[seq(1,111,by=3)]
> persp(age, year, log(usa1950$rate$male[seq(1,111,by=3), seq(1,62,by=3)]), theta=-30,
+ main="Observed death rates", col=grey(.93), shade=TRUE, xlab="Age", ylab="",
+ zlab="Log death rate", ticktype="detailed", cex.axis=0.6, cex.lab=0.6)
> persp(age, year, predict(fitBIC)[seq(1,111,by=3),seq(1,62,by=3)], theta=-30,
+ main="Smoothed death rates", col=grey(.93), shade=TRUE, xlab="Age", ylab="",
+ zlab="Log death rate", ticktype="detailed", cex.axis=0.6, cex.lab=0.6)
```

Figure 8.4 shows smoothing of the 1950–2010 dataset. On the left are observed death rates, $m_{x,t}$, and on the right are the estimated smoothed rates, $f_t(x)$.

8.3 Lee–Carter and Related Forecasting Methods

The Lee–Carter (LC) method introduced in Lee & Carter (1992) for forecasting mortality rates uses principal components analysis to decompose the age-time matrix of log central

**FIGURE 8.4**

Smoothed male mortality rates using bivariate P-splines, United States, 1950–2010.

death rates into a linear combination of age and time parameters. The time parameter is used in forecasting.

LC has spawned numerous variants and extensions. The two main variants of LC are Lee-Miller (LM) (Lee & Miller 2001) and Booth-Maindonald-Smith (BMS) (Booth et al. 2002). Others result from different combinations of possible options. These variants are collectively referred to as “LC methods”. A major extension of this approach uses functional data models (FDM); first proposed by Hyndman & Ullah (2007), it was further developed by Hyndman & Booth (2008) and Hyndman & Shang (2009). Again, various combinations of options produce variations within the collectively labeled “HU methods”.

We identify six methods by their proponents; these are listed in Table 8.1 where the defining features of the models are shown. Most authors referring to the “Lee-Carter method” actually refer to the generic model in which all available data are used, there is no adjustment of the time parameter prior to forecasting, and fitted rates are used as jump-off rates; Booth et al. (2006) labeled this “LCnone”. Note that within the options listed in Table 8.1 there are twenty-four possible combinations (4 adjustment options $\times$ 3 data period options $\times$ 2 jump-off options) for the LC methods. For the HU methods, additional options have been defined by varying the data period option to include 1950 (Shang et al. 2011). Clearly, any date can be used for the start of the data period.

In this section, we begin with the complete available dataset `usa`, for which ages range from 0 to 110+ and years range from 1933 to 2010.

```
> usa
Mortality data for USA
Series: female male total
Years: 1933 - 2010
Ages: 0 - 110
```

Appropriate age ranges and fitting periods are defined for each method.

As the Lee-Carter methods do not incorporate smoothing, it is advisable to avoid erroneous rates at the oldest ages when using these methods.

TABLE 8.1

Lee–Carter and Hyndman–Ullah methods by defining features.

Method	Data Period	Smoothing	Adjustment to Match	Jump-off Rates	Reference
Lee–Carter Methods					
LC	All	No	D_t	Fitted	Lee & Carter (1992)
LM	1950	No	$e(0)$	Observed	Lee & Miller (2001)
BMS	Linear	No	$D_{x,t}$	Fitted	Booth et al. (2002)
LCnone	All	No	–	Fitted	–
Hyndman–Ullah Methods					
HU	–	Yes	–	Fitted	Hyndman & Ullah (2007)
HUrob	–	Yes	–	Fitted	Hyndman & Ullah (2007)
HUw	–	Yes	–	Fitted	Hyndman & Shang (2009)

```
> usa.90 <- extract.ages(usa, ages=0:90)
```

In this example, the upper age group is then 90+.

8.3.1 Lee–Carter (LC) Method

The model structure proposed by Lee & Carter (1992) is given by

$$\log(m_{x,t}) = a_x + b_x k_t + \varepsilon_{x,t}, \quad (8.1)$$

where a_x is the age pattern of the log mortality rates averaged across years, b_x is the first principal component reflecting relative change in the log mortality rate at each age, k_t is the first set of principal component scores by year t and measures the general level of the log mortality rates, and $\varepsilon_{x,t}$ is the residual at age x and year t . The model assumes homoskedastic error and is estimated using a singular value decomposition.

The LC model in Equation (8.1) is over-parameterized in the sense that the model structure is invariant under the following transformations:

$$\begin{aligned} \{a_x, b_x, k_t\} &\mapsto \{a_x, b_x/c, ck_t\}, \\ \{a_x, b_x, k_t\} &\mapsto \{a_x - cb_x, b_x, k_t + c\}. \end{aligned}$$

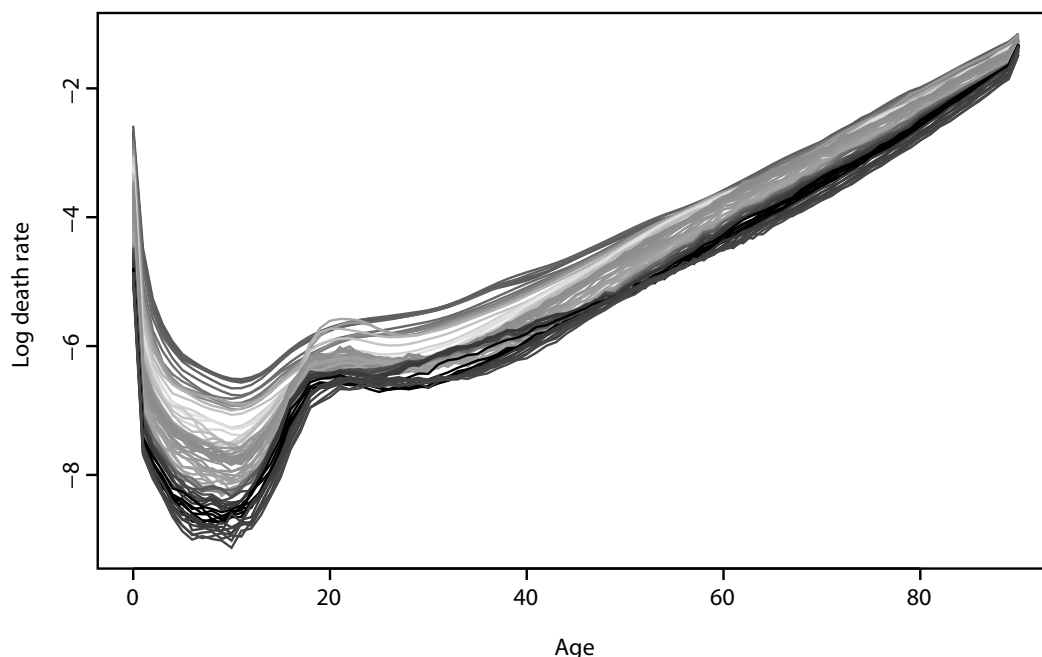
In order to ensure the model's identifiability, Lee & Carter (1992) imposed two constraints, given as

$$\sum_{t=1}^n k_t = 0, \quad \sum_{x=x_1}^{x_p} b_x = 1.$$

In addition, the LC method adjusts k_t by refitting to the total number of deaths. This adjustment gives more weight to high rates, thus roughly counterbalancing the effect of using a log transformation of the mortality rates. The adjusted k_t is then extrapolated using ARIMA models. Lee & Carter (1992) used a random walk with drift (RWD) model, which can be expressed as

$$k_t = k_{t-1} + d + e_t,$$

where d is known as the drift parameter and measures the average annual change in the series, and e_t is an uncorrelated error. It is notable that the RWD model provides satisfactory results in many cases (Tuljapurkar, Li & Boe 2000, Lee & Miller 2001, Lazar & Denuit

**FIGURE 8.5**

U.S. male mortality rates, 1933–2010. Note the emergence of the “accident hump” at about age 20 and the effect of deaths due to AIDS at about ages 25–44 in 1985–1995.

2009). From this forecast of the principal component scores, the forecast age-specific log mortality rates are obtained using the estimated age effects a_x and b_x , and setting $\varepsilon_{x,t} = 0$, in Equation (8.1).

The LC method is implemented in the R **demography** package as follows:

```
> lc.male <- lca(usa.90, series="male")
> forecast.lc.male <- forecast(lc.male, h=20)
```

The estimated age parameters a_x and b_x , and the estimated time parameter k_t are respectively obtained using `lc.male$ax`, `lc.male$bx` and `lc.male$kt`. Similarly, the forecast time parameter (rescaled to zero in the jump-off year 2010) is obtained using `forecast.lc.male$kt`.

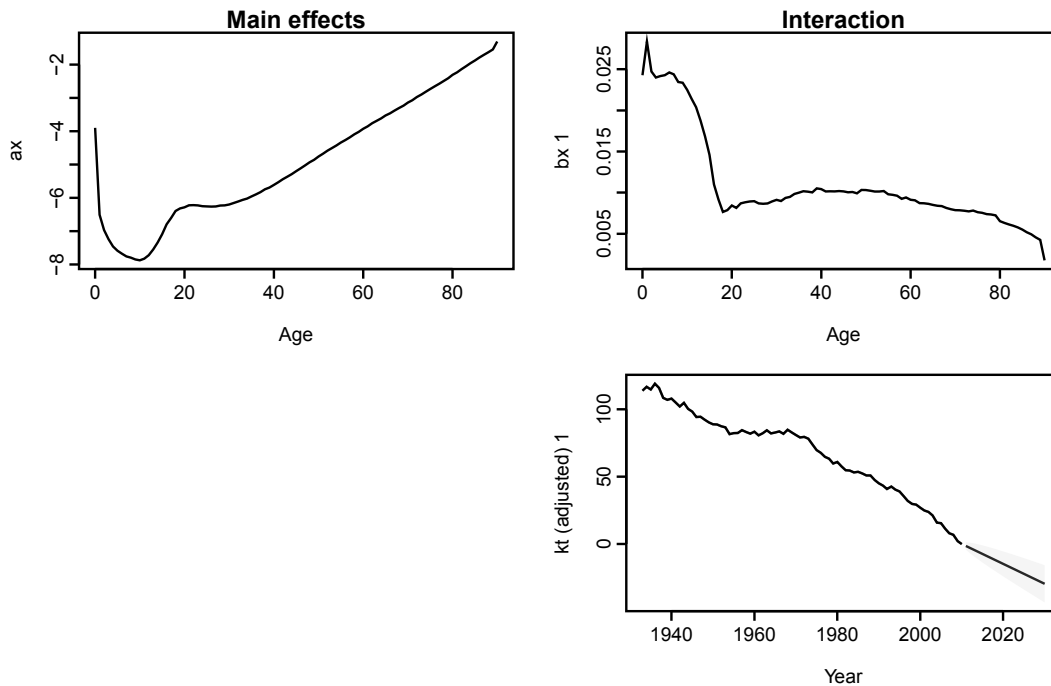
The data (Figure 8.5), model parameters (Figure 8.6) and forecasts can be viewed via

```
> plot(usa.90, series="male")
> plot(lc.male)
> plot(forecast.lc.male, plot.type="component")
> plot(usa.90, series="male", ylim=c(-10,0), lty=2)
> lines(forecast.lc.male)
```

The LC method without adjustment of k_t (LCnone) is achieved by choosing the adjustment option `adjust="none"`.

```
> lcnone.male <- lca(usa.90, series="male", adjust="none")
```

The effect of the LC adjustment of k_t is seen in Figure 8.7 via

**FIGURE 8.6**

LC model and forecast, U.S. male mortality. Fitting period = 1933–2010; forecasting horizon = 20 years.

```
> plot(lcnone.male$kt, ylab="kt",ylim=c(-70,90), xlab="")
> lines(lc.male$kt, lty=2)
> legend("topright", lty=1:2, legend=c("LCnone","LC"))
```

Differences between these two lines in the first and last years of the fitting period can have a substantial effect on the forecast.

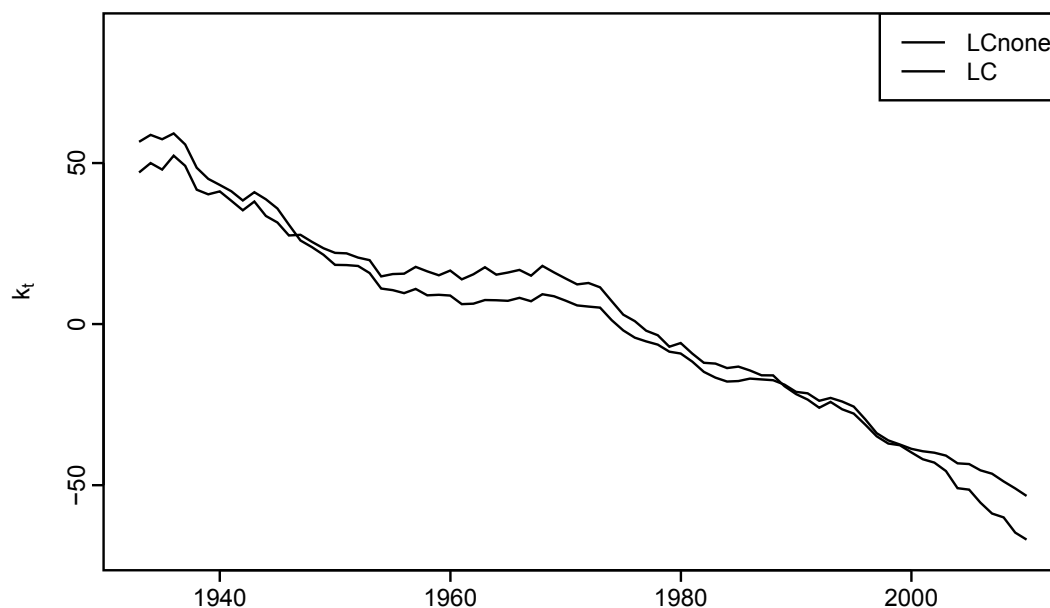
An alternative, and more efficient, approach to estimating a Lee–Carter model was described by Brouhns et al. (2002); it involves embedding the method in a Poisson regression model, and using maximum likelihood estimation. This can be achieved in R using, for example,

```
> lca(usa, series="male", adjust="dxt")
```

8.3.2 Lee–Miller (LM) Method

The LM method is a variant of the LC method. It differs from the LC method in three ways:

- The fitting period begins in 1950;
- The adjustment of k_t involves fitting to the life expectancy $e(0)$ in year t ;
- The jump-off rates are the observed rates in the jump-off year instead of the fitted rates.

**FIGURE 8.7**

LC k_t with and without adjustment, U.S. male mortality, 1933–2010.

In their evaluation of the LC method, Lee & Miller (2001) found that the pattern of change in mortality rates was not constant over time, which is a strong assumption of the LC method. Consequently, the adjustment of historical principal component scores resulted in a large estimation error. To overcome this, Lee & Miller (2001) adopted 1950 as the commencing year of the fitting period due to different age patterns of change for 1900–1949 and 1950–1995. This fitting period had previously been used by Tuljapurkar et al. (2000).

In addition, the adjustment of k_t was done by fitting to observed life expectancy in year t , rather than by fitting to total deaths in year t . This has the advantage of eliminating the need for population data. Further, Lee & Miller (2001) found a mismatch between fitted rates for the final year of the fitting period and observed rates in that year. This jump-off error was eliminated by using observed rates in the jump-off year.

The LM method is implemented as follows:

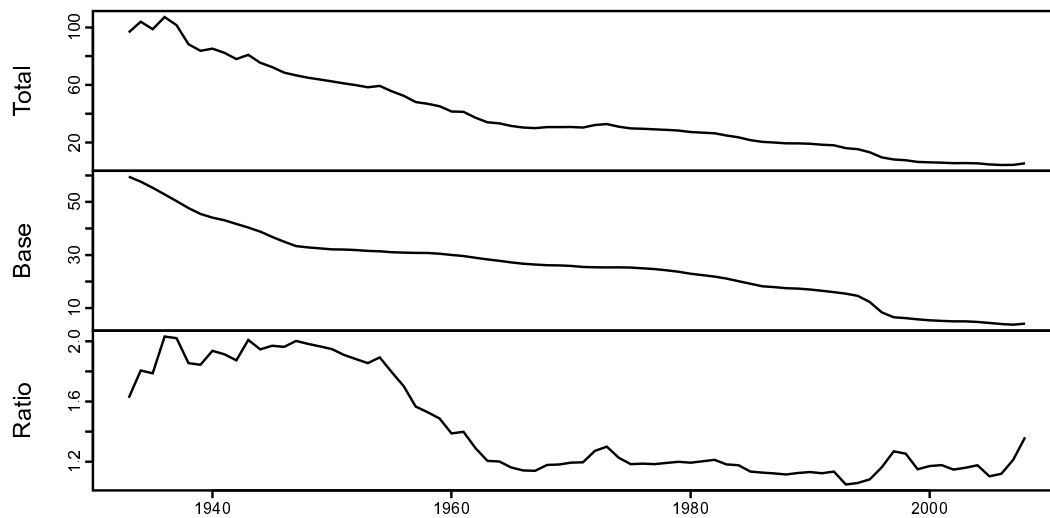
```
> lm.male <- lca(usa.90, series="male", adjust="e0", years=1950:max(usa$year))
> forecast.lm.male <- forecast(lm.male, h=20, jumpchoice = "actual")
```

The LM method has been found to produce more accurate forecasts than the original LC method (Booth et al. 2005, 2006).

8.3.3 Booth–Maindonald–Smith (BMS) Method

The BMS method is another variant of the LC method. The BMS method differs from the LC method in three ways:

- The fitting period is determined on the basis of a statistical ‘goodness of fit’ criterion, under the assumption that the principal component score k_1 is linear;
- The adjustment of k_t involves fitting to the age distribution of deaths rather than to the total number of deaths;

**FIGURE 8.8**

Mean deviances for base and total models and their ratio, U.S. male mortality, 1933–2010.

- The jump-off rates are the fitted rates under this fitting regime.

A common feature of the LC method is the linearity of the best fitting time series model of the first principal component score, but Booth, Maindonald & Smith (2002) found the linear time series to be compromised by structural change. By first assuming the linearity of the first principal component score, the BMS method seeks to achieve the optimal ‘goodness of fit’ by selecting the optimal fitting period from all possible fitting periods ending in year n . The optimal fitting period is determined based on the smallest ratio of the mean deviances of the fit of the underlying LC model to the overall linear fit.

Instead of fitting to the total number of deaths, the BMS method uses a quasi-maximum likelihood approach by fitting the Poisson distribution to model age-specific deaths, and using deviance statistics to measure the ‘goodness of fit’ (Booth, Maindonald & Smith 2002). The jump-off rates are taken to be the fitted rates under this adjustment.

The BMS method is implemented thus:

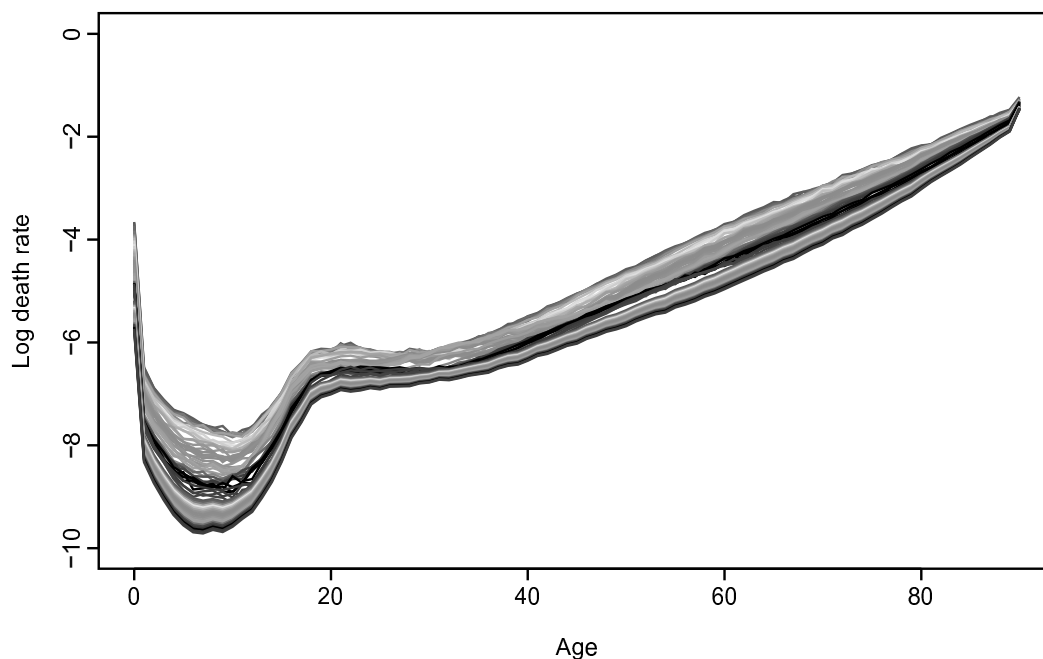
```
> bms.male <- bms(usa.90 series="male", minperiod = 30, breakmethod = "bms")
> forecast.bms.male <- forecast(bms.male, h=20)
```

It is advisable to review the deviances (Figure 8.8) and the automatically chosen fitting period, which is based on a local minimum of the ratio of total to base mean deviances.

```
> plot(bms.male$mdevs, main="Mean deviances for base and total models", xlab="")
> bms.male$year[1]
[1] 1967
```

Estimated k_t and forecast rates (Figure 8.9) are obtained via

```
> plot(bms.male$kt)
> plot(usa.90, series="male", years=bms.male$year[1]:max(usa$year),
+ ylim=c(-10,0), lty=2, main="BMS method: observed (1979–2010)
+ and forecast (2011–2030) rates")
> lines(forecast.bms.male)
```

**FIGURE 8.9**

Observed (1979–2010) and forecast (2011–2030) mortality rates using the BMS method for U.S. males.

An alternative implementation using the `lca()` function, which permits all possible variants to be produced, is:

```
> bms.male <- lca(usa.90, series="male", adjust="dxt", chooseperiod=TRUE,
+ minperiod = 30, breakmethod = "bms")
> forecast.bms.male <- forecast(bms.male, h=20)
```

Forecasts from the BMS method have been found to be more accurate than those from the original LC method and of similar accuracy as those from the LM method (Booth et al. 2005, 2006).

8.3.4 Hyndman–Ullah (HU) Method

Using the functional data analysis paradigm of Ramsay & Silverman (2005), Hyndman & Ullah (2007) proposed a nonparametric method for modeling and forecasting log mortality rates. This approach extends the LC method in four ways:

- The log mortality rates are smoothed prior to modeling;
- Functional principal components analysis is used;
- More than one principal component is used in forecasting;
- The forecasting models for the principal component scores are typically more complex than the RWD model.

The log mortality rates are smoothed using penalized regression splines as described in Section 8.2. To emphasize that age, x , is now considered a continuous variable, we write $m_t(x)$ to represent mortality rates for age $x \in [x_1, x_p]$ in year t . We then define $z_t(x) = \log m_t(x)$ and write

$$z_t(x_i) = f_t(x_i) + \sigma_t(x_i)\varepsilon_{t,i}, \quad i = 1, \dots, p, \quad t = 1, \dots, n, \quad (8.2)$$

where $f_t(x_i)$ denotes a smooth function of x as before; $\sigma_t(x_i)$ allows the amount of noise to vary with x_i in year t , thus rectifying the assumption of homoskedastic error in the LC model; and $\varepsilon_{t,i}$ is an independent and identically distributed standard normal random variable.

Given continuous age x , functional principal components analysis (FPCA) is used in the decomposition. The set of age-specific mortality curves is decomposed into orthogonal functional principal components and their uncorrelated principal component scores. That is,

$$f_t(x) = a(x) + \sum_{j=1}^J b_j(x)k_{t,j} + e_t(x), \quad (8.3)$$

where $a(x)$ is the mean function estimated by $\hat{a}(x) = \frac{1}{n} \sum_{t=1}^n f_t(x)$; $\{b_1(x), \dots, b_J(x)\}$ is a set of the first J functional principal components; $\{k_{t,1}, \dots, k_{t,J}\}$ is a set of uncorrelated principal component scores; $e_t(x)$ is the residual function with mean zero; and $J < n$ is the number of principal components used. Note that we use $a(x)$ rather than a_x to emphasise that x is not treated as a continuous variable.

Multiple principal components are used because the additional components capture non-random patterns that are not explained by the first principal component (Booth, Maindonald & Smith 2002, Renshaw & Haberman 2003, Koissi, Shapiro & Högnäs 2006). Hyndman & Ullah (2007) found $J = 6$ to be larger than the number of components actually required to produce white noise residuals, and this is the default value. The conditions for the existence and uniqueness of $k_{t,j}$ are discussed by Cardot, Ferraty & Sarda (2003).

Although Lee & Carter (1992) did not rule out the possibility of a more complex time series model for the k_t series, in practice an RWD model has typically been employed in the LC method. For higher order principal components, which are orthogonal by definition to the first component, other time series models arise for the principal component scores. For all components, the HU method selects the optimal time series model using standard model-selection procedures (e.g. AIC). By conditioning on the observed data $\mathcal{I} = \{z_1(x), \dots, z_n(x)\}$ and the set of functional principal components $\mathbf{B} = \{b_1(x), \dots, b_J(x)\}$, the h -step-ahead forecast of $z_{n+h}(x)$ can be obtained by

$$\hat{z}_{n+h|n}(x) = \mathbb{E}[z_{n+h}(x)|\mathcal{I}, \mathbf{B}] = \hat{a}(x) + \sum_{j=1}^J b_j(x)\hat{k}_{n+h|n,j},$$

where $\hat{k}_{n+h|n,j}$ denotes the h -step-ahead forecast of $k_{n+h,j}$ using a univariate time series model, such as the optimal ARIMA model selected by the automatic algorithm of Hyndman & Khandakar (2008), or an exponential smoothing state space model (Hyndman et al. 2008).

Because of the orthogonality of all components, it is easy to derive the forecast variance as

$$\hat{v}_{n+h|n}(x) = \text{var}[z_{n+h}(x)|\mathcal{I}, \mathbf{B}] = \sigma_a^2(x) + \sum_{j=1}^J b_j^2(x)u_{n+h|n,j} + v(x) + \sigma_t^2(x),$$

where σ_a^2 is the variance of $\hat{a}(x)$; $u_{n+h|n,j}$ is the variance of $k_{n+h,j} | k_{1,j}, \dots, k_{n,j}$ (obtained from the time series model); $v(x)$ is the variance of $e_t(x)$ and $\sigma_t(x)$ is defined in (8.2). This expression is used to construct prediction intervals for future mortality rates in R.

For the Hyndman–Ullah methods, we return to the complete dataset `usa`.

```
> usa
Mortality data for USA
  Series: female male total
  Years: 1933 - 2010
  Ages: 0 - 110
```

In order to avoid the war years, the fitting period commences in 1950. Given that smoothing is employed, the age range is 0 to 100+.

```
> usa1950.100 <- extract.years(extract.ages(usa, 0:100), years=1950:max(usa$year))
> usa1950.100
Mortality data for USA
  Series: female male total
  Years: 1950 - 2010
  Ages: 0 - 100
```

The HU method is implemented as below. The model and forecast are seen in Figures 8.10 and 8.11.

```
> smus1950.100 <- smooth.demogdata(usa1950.100)
> fdm.male <- fdm(smus1950.100, series="male", order=3)
> forecast.fdm.male <- forecast.fdm(fdm.male, h=20)
> plot(forecast.fdm.male, plot.type="component")
> plot(smus1950.100, series="male", ylim=c(-10,0), lty=2, main ="HU method:
+ observed (1950–2010) and forecast (2011–2030) rates")
> lines(forecast.fdm.male)
```

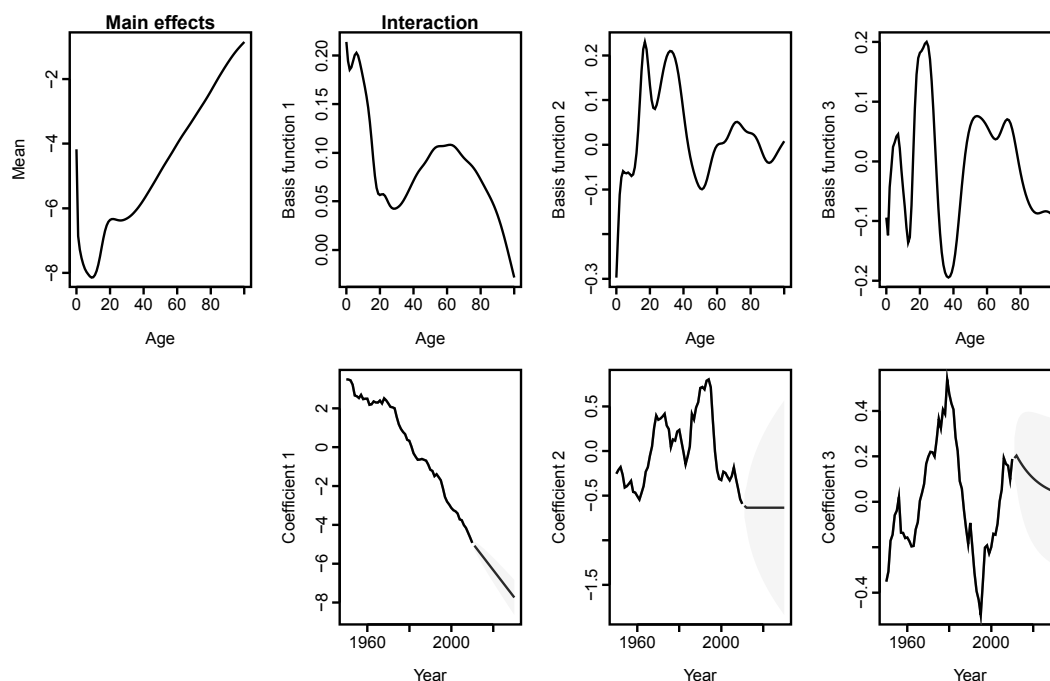
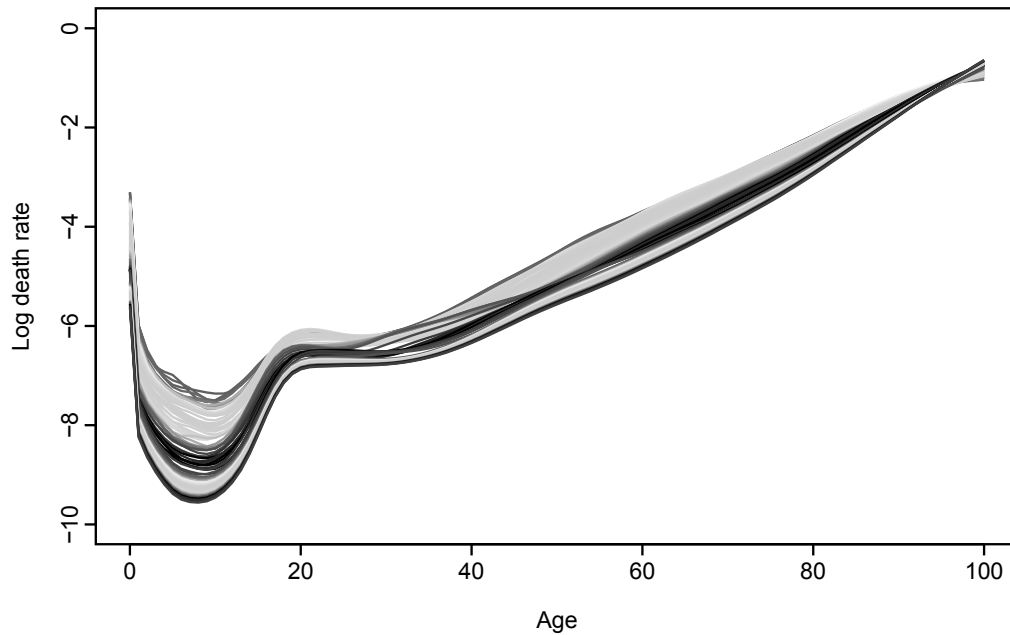


FIGURE 8.10

HU model and forecast, U.S. male mortality. Fitting period = 1950–2010; forecasting horizon = 20 years.

**FIGURE 8.11**

Observed (1950-2000) and forecast (2011-2030) mortality rates using the HU method for U.S. males.

8.3.5 Robust Hyndman–Ullah (HUrob) Method

The presence of outliers can seriously affect the performance of modelling and forecasting. The HUrob method is designed to eliminate their effect. This method utilizes the reflection-based principal component analysis (RAPCA) algorithm of Hubert, Rousseeuw & Verboven (2002) to obtain projection-pursuit estimates of principal components and their associated scores. The integrated squared error provides a measure of the accuracy of the principal component approximation for each year (Hyndman & Ullah 2007). Outlying years would result in a larger integrated squared error than the critical value obtained by assuming normality of $e_t(x)$ (see Hyndman & Ullah 2007, for details). By assigning zero weight to outliers, the HU method can then be used to model and forecast mortality rates without the possible influence of outliers.

The HUrob method is implemented as follows:

```
> fdm.male <- fdm(smus1950.100, series="male", method="rapca")
> forecast.fdm.male <- forecast.fdm(fdm.male, h=20)
> plot(smus1950.100, series="male", ylim=c(-10,0), lty=2, main ="HUrob method:
+ observed (1950-2010) and forecast (2011-2030) rates")
> lines(forecast.fdm.male)
```


8.3.6 Weighted Hyndman–Ullah (HUw) Method

The HU method does not weight annual mortality curves in the functional principal components analysis. However, it might be argued that more recent experience has greater relevance to the future than more distant experience. The HUw method uses geometrically decaying weights in the estimation of the functional principal components, thus allowing these quantities to be based more on recent data than on data from the distant past.

The weighted functional mean $\hat{a}^*(x)$ is estimated by the weighted average

$$\hat{a}^*(x) = \sum_{t=1}^n w_t f_t(x), \quad (8.4)$$

where $\{w_t = \beta(1 - \beta)^{n-t}, t = 1, \dots, n\}$ denotes a set of weights, and $0 < \beta < 1$ denotes the weight parameter. Hyndman & Shang (2009) describe how to estimate β from the data. The set of weighted curves $\{w_t[f_t(x) - \hat{a}^*(x)]; t = 1, \dots, n\}$ is decomposed using FPCA:

$$f_t(x) = \hat{a}^*(x) + \sum_{j=1}^J b_j^*(x) k_{t,j} + e_t(x), \quad (8.5)$$

where $\{b_1^*(x), \dots, b_J^*(x)\}$ is a set of weighted functional principal components. By conditioning on the observed data $\mathcal{I} = \{z_1(x), \dots, z_n(x)\}$ and the set of weighted functional principal components $\mathbf{B}^*$, the h -step-ahead forecast of $z_{n+h}(x)$ can be obtained by

$$\hat{z}_{n+h|n}(x) = \mathbb{E}[z_{n+h}(x) | \mathcal{I}, \mathbf{B}^*] = \hat{a}^*(x) + \sum_{j=1}^J b_j^*(x) \hat{k}_{n+h|n,j}.$$

The HUw method is implemented as follows:

```
> fdm.male <- fdm(smus1950.100, series="male", method="classical", weight=TRUE, beta=0.1)
> forecast.fdm.male <- forecast.fdm(fdm.male, h=20)
> plot(smus1950.100, series="male", ylim=c(-10,0), lty=2, main = "HUw method:
+ observed (1950-2010) and forecast (2011-2030) rates")
> lines(forecast.fdm.male)
```

8.4 Other Mortality Forecasting Methods

Other extrapolative mortality forecasting methods are included here for completeness, but are not considered in detail as the methods are not fully implemented in packages available on CRAN.

A number of methods have been developed to account for the significant impact of cohort (year of birth) in some countries. In the United Kingdom, males born around 1931 have experienced higher rates of mortality improvement than earlier or later cohorts (Willets 2004); less marked effects have also been observed elsewhere (Cairns 2009).

The Renshaw and Haberman (RH) (2006) extension to Lee-Carter to include cohort effects can be written as¹

$$\log(m_{x,t}) = a_x + b_x^1 k_t + b_x^2 \gamma_{t-x} + \varepsilon_{x,t}, \quad (8.6)$$

¹For clarity, models have been written in a standardised format which may in some cases differ from the form used by the authors originally. a_x and b_x terms are used for age-related effects, k_t terms for period-related effects, and γ_{t-x} terms for cohort-related effects. Models originally expressed in terms of the force of mortality are expressed in terms of the central death rate; these are equivalent under the assumption of a constant force of mortality over each year of age.

where a_x is the age pattern of the log mortality rates averaged across years, k_t represents the general level of mortality in year t , γ_{t-x} represents the general level of mortality for the cohort born in year $(t-x)$, b_x^1 and b_x^2 measure the relative response at age x to changes in k_t and γ_{t-x} , respectively, and $\varepsilon_{x,t}$ is the residual at age x . The fitted k_t and γ_{t-x} parameters are forecast using univariate time series models. The model can be implemented using the **ilc** functions (Butt & Haberman 2009).

A subsequent related model in which b_x^2 is set equal to 1 at all ages (Haberman & Renshaw 2011) was found to resolve some forecasting issues associated with the original. The Age-Period-Cohort (APC) model (Currie 2006) incorporates age, time and cohort effects that are independent in their effects on mortality,

$$\log(m_{x,t}) = a_x + k_t + \gamma_{t-x} + \varepsilon_{x,t}. \quad (8.7)$$

The two-dimensional P-spline method of Currie et al. (2004) has already been described in Section 8.2. Forecast rates are estimated simultaneously with fitting the mortality surface. Implementation of the two-dimensional P-spline method to produce mortality forecasts uses the **MortalitySmooth** package. Forecasts of U.S. male mortality rates and plots of age 65 and age 85 forecast rates with prediction intervals can be produced as follows, following the commands already shown in Section 8.2:

```
> library(MortalitySmooth)
> forecastyears <- 2011:2030
> forecastdata <- list(x=usa1950$age, y=forecastyears)
> CDEpredict <- predict(fitBIC, newdata=forecastdata, se.fit=TRUE)
> whiA <- c(66,86)
> plot(usa1950, series="male", age=whiA-1, plot.type="time",
+ xlim=c(1950,2030), ylim=c(-6.2,-1), xlab="years",
+ main="USA: male projected death rates using 2-dimensional CDE method", col=c(1,2))
> matlines(forecastyears, t(CDEpredict$fit[whiA,]), lty=1, lwd=2)
> matlines(forecastyears, t(CDEpredict$fit[whiA,]+2*CDEpredict$se.fit[whiA,]), lty=2)
> matlines(forecastyears, t(CDEpredict$fit[whiA,]-2*CDEpredict$se.fit[whiA,]), lty=2)
> legend("bottomleft", lty=1, col=1:2, legend=c("Age 65", "Age 85"))
```

In addition to being applied in the age and period dimensions, the two-dimensional P-spline method can incorporate cohort effects by instead being applied to age-cohort data.

Cairns et al. (2006a) have forecast mortality at older ages using a number of models for $\text{logit}(q_{x,t}) = \log[q_{x,t}/(1 - q_{x,t})]$, where $q_{x,t}$ is the probability that an individual aged x at time t will die before time $t + 1$. The original CBD model (Cairns et al. 2006a) is

$$\text{logit}(q_{x,t}) = k_t^1 + (x - \bar{x})k_t^2 + \varepsilon_{x,t}, \quad (8.8)$$

where $\bar{x}$ is the mean age in the sample range. Later models (Cairns 2009) incorporate a combination of cohort effects and a quadratic term for age:

$$\text{logit}(q_{x,t}) = k_t^1 + (x - \bar{x})k_t^2 + \gamma_{t-x} + \varepsilon_{x,t}, \quad (8.9)$$

$$\text{logit}(q_{x,t}) = k_t^1 + (x - \bar{x})k_t^2 + ((x - \bar{x})^2 - \hat{\sigma}_x^2) + \gamma_{t-x} + \varepsilon_{x,t}, \quad (8.10)$$

$$\text{logit}(q_{x,t}) = k_t^1 + (x - \bar{x})k_t^2 + (x_c - x)\gamma_{t-x} + \varepsilon_{x,t}, \quad (8.11)$$

where the constant parameter x_c is to be estimated and the constant $\hat{\sigma}_x^2$ is the mean of $(x - \bar{x})^2$. Other authors (e.g., Plat 2009) have proposed related models.

The **LifeMetrics** R software package implements the Lee-Carter method (using maximum likelihood estimation and a Poisson distribution for deaths) along with RH, APC, P-splines and the four CBD methods. The software, which is not part of CRAN, is available from www.life-metrics.com.

jpmorgan.com/pages/jpmorgan/investbk/solutions/lifemetrics/software. The software and the methods it implements is described in detail in Coughlan et al. (2007).

De Jong & Tickle (2006) (DJT) tailor the state space framework to create a method that integrates model estimation and forecasting, while using B-splines to reduce dimensionality and build in the expected smooth behaviour of mortality over age. Compared with Lee-Carter, the method uses fewer parameters, produces smooth forecast rates and offers the advantages of integrated estimation and forecasting. A multi-country evaluation of out-of-sample forecast performance found that LM, BMS, HU and DJT gave significantly more accurate forecast log mortality rates relative to the original LC, with no one method significantly more accurate than the others (Booth et al. 2006).

8.5 Coherent Mortality Forecasting

In modelling mortality for two or more sub-populations of a larger population simultaneously, it is usually desirable that the forecasts are non-divergent or “coherent”. The Product-Ratio method (Hyndman et al. 2013) achieves coherence through the convergence to a set of appropriate constants of forecast age-specific ratios of death rates for any two sub-populations. The method makes use of functional forecasting (HU methods).

The method is presented here in terms of forecasting male and female age-specific death rates; extension to more than two sub-populations is straightforward (Hyndman et al. 2013). Let $s_{t,F}(x) = \exp[f_{t,F}(x)]$ denote the smoothed female death rate for age x and year t , $t = 1, \dots, n$. Similar notation applies for males.

Let the square roots of the products and ratios of the smoothed rates for each sex be

$$p_t(x) = \sqrt{s_{t,M}(x)s_{t,F}(x)} \quad \text{and} \quad r_t(x) = \sqrt{s_{t,M}(x)/s_{t,F}(x)}.$$

These are modeled by functional time series models:

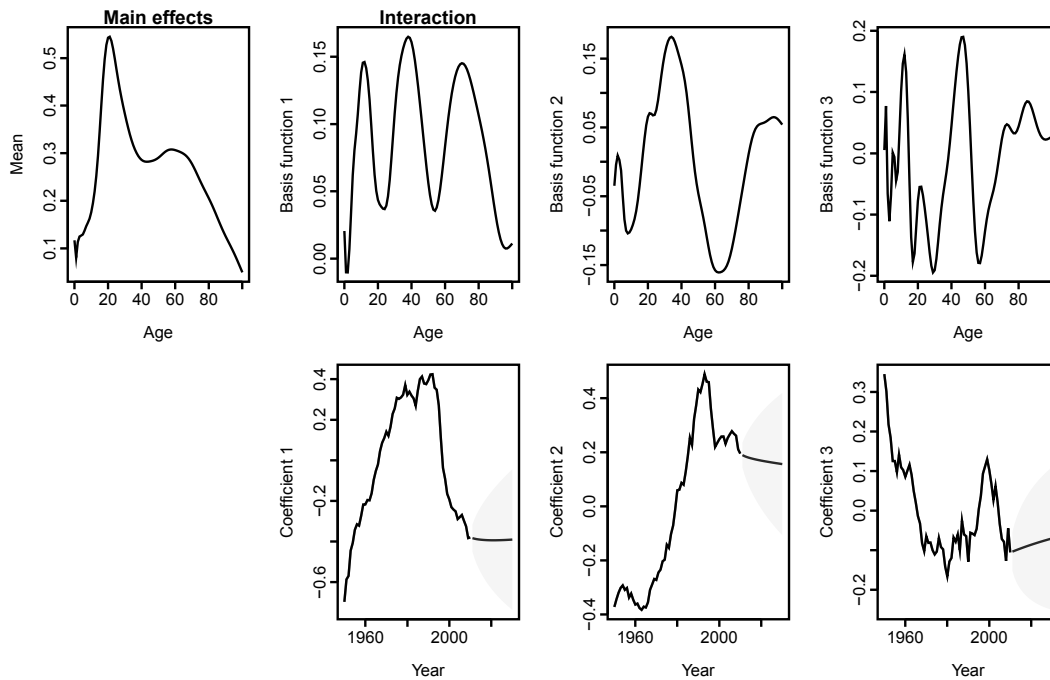
$$\log[p_t(x)] = \mu_p(x) + \sum_{k=1}^K \beta_{t,k} \phi_k(x) + e_t(x), \quad (8.12a)$$

$$\log[r_t(x)] = \mu_r(x) + \sum_{\ell=1}^L \gamma_{t,\ell} \psi_\ell(x) + w_t(x), \quad (8.12b)$$

where the functions $\{\phi_k(x)\}$ and $\{\psi_\ell(x)\}$ are the principal components obtained from decomposing $\{p_t(x)\}$ and $\{r_t(x)\}$, respectively, and $\beta_{t,k}$ and $\gamma_{t,\ell}$ are the corresponding principal component scores. The function $\mu_p(x)$ is the mean of the set of curves $\{p_t(x)\}$, and $\mu_r(x)$ is the mean of $\{r_t(x)\}$. The error terms, given by $e_t(x)$ and $w_t(x)$, have zero mean and are serially uncorrelated.

The coefficients $\{\beta_{t,1}, \dots, \beta_{t,K}\}$ and $\{\gamma_{t,1}, \dots, \gamma_{t,L}\}$ are forecast using time series models, as detailed in Section 8.3.4. To ensure the forecasts are coherent, the coefficients $\{\gamma_{t,\ell}\}$ are constrained to be stationary processes. The forecast coefficients are then multiplied by the basis functions, resulting in forecasts of the curves $p_t(x)$ and $r_t(x)$ for future t . If $p_{n+h|n}(x)$ and $r_{n+h|n}(x)$ are h -step forecasts of the product and ratio functions, respectively, then forecasts of the sex-specific death rates are obtained using $s_{n+h|n,M}(x) = p_{n+h|n}(x)r_{n+h|n}(x)$ and $s_{n+h|n,F}(x) = p_{n+h|n}(x)/r_{n+h|n}(x)$.

The method makes use of the fact that the product and ratio behave roughly independently of each other provided the sub-populations have approximately equal variances. (If

**FIGURE 8.12**

Ratio function decomposition (1950–2010) with forecast (2011–2030), U.S. mortality.

there are substantial differences in the variances, the forecasts remain unbiased but less efficient.)

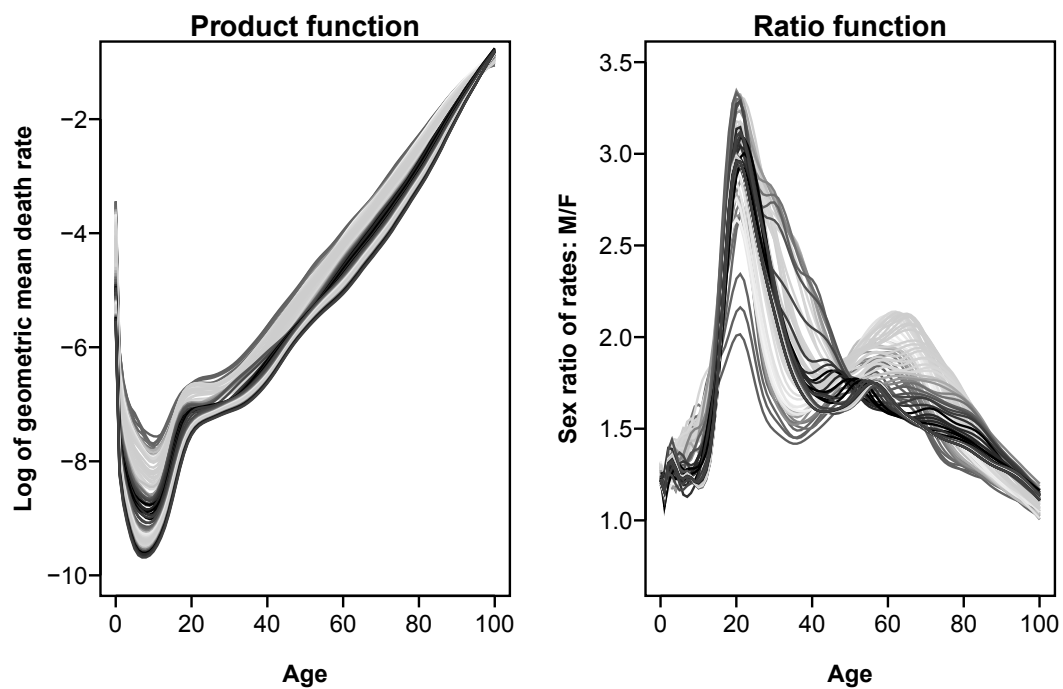
The Product-Ratio method is illustrated in Figures 8.12 to 8.14 and implemented as follows:

```
> usa.pr <- coherentfcm(smus1950.100, weight=TRUE, beta=0.05)
> usa.pr.f <- forecast(usa.pr, h=20)

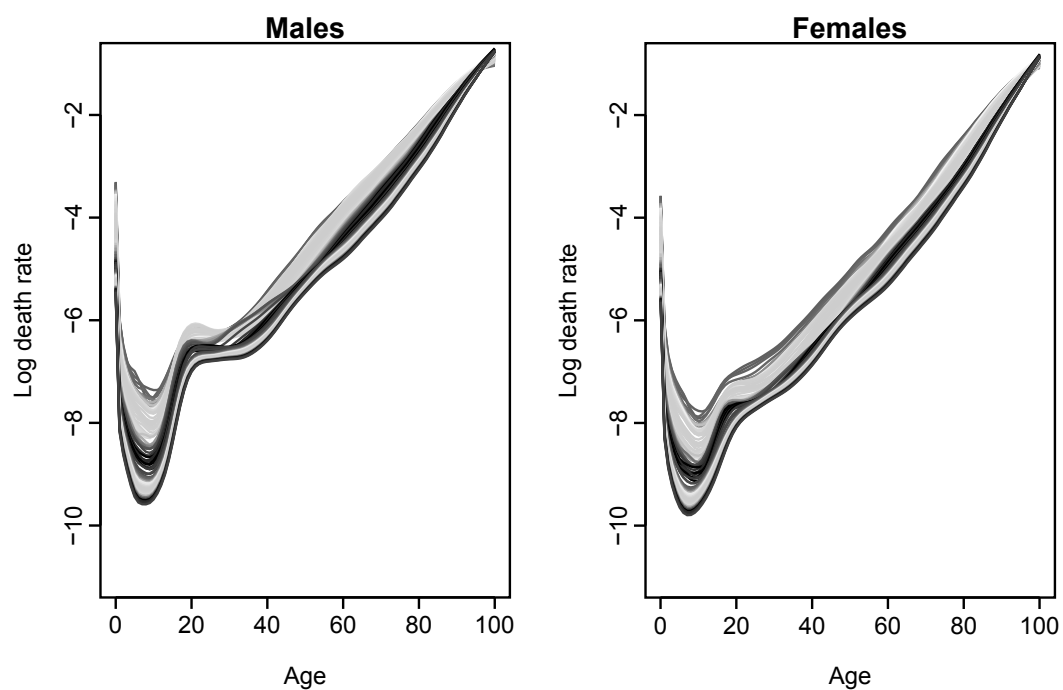
> plot(usa.pr.f$product, plot.type="component", components=3)
> plot(usa.pr.f$ratio$male, plot.type="component", components=3)

> par(mfrow=c(1,2))
> plot(usa.pr$product$y, ylab="Log of geometric mean death rate", font.lab=2,
+ lty=2, las=1, ylim=c(-10,-1), main="Product function")
> lines(usa.pr.f$product)
> plot(sex.ratio(smus1950.100), ylab="Sex ratio of rates: M/F", ylim=c(0.7,3.5),
+ lty=2, las=1, font.lab=2, main="Ratio function")
> lines(sex.ratio(usa.pr.f))

> plot(smus1950.100, series="male", lty=2, ylim=c(-11,-1), main="Males")
> lines(usa.pr.f$male)
> plot(smus1950.100, series="female", lty=2, ylim=c(-11,-1), main="Females")
> lines(usa.pr.f$female)
```

**FIGURE 8.13**

Observed(1950–2010) and forecast(2011–2030) product and ratio functions, U.S. mortality.

**FIGURE 8.14**

Observed(1950–2010) and forecast(2011–2030) male and female mortality rates using the product-ratio method with functional data models, United States.

8.6 Life Table Forecasting

The methods described in this chapter generate forecast m_x rates, which can then be used to produce forecast life table functions using standard methods (e.g. Chiang 1984). Assuming that m_x rates are available for ages $0, 1, \dots, \omega - 1, \omega +$, the `lifetable` function in `demography` generates life table functions from a radix of $l_0 = 1$ as follows for single years of age to $\omega - 1$:

$$q_x = m_x / (1 + (1 - a_x)m_x), \quad (8.13)$$

$$d_x = l_x q_x, \quad (8.14)$$

$$l_{x+1} = l_x - d_x, \quad (8.15)$$

$$L_x = l_x - d_x(1 - a_x), \quad (8.16)$$

$$T_x = L_x + L_{x+1} + \dots + L_{\omega-1} + L_{\omega+}, \quad (8.17)$$

$$e_x = T_x / l_x, \quad (8.18)$$

where $a_x = 0.5$ for $x = 1, \dots, \omega - 1$, and a_0 values (which allow for the fact that deaths in this age group occur earlier than midway through the year of age on average) are from Coale et al. (1983). For the final age group, $q_{\omega+} = 1$, $L_{\omega+} = l_{\omega}/m_{\omega}$, and $T_{\omega+} = L_{\omega+}$. For life tables commencing at an age other than zero, the same formulae apply, generated from a radix of 1 at the commencing age.

The `demography` package produces life tables using `lifetable`, and life expectancies using the function `life.expectancy`. For forecast life expectancies, `flife.expectancy` is used to produce the point forecast and the prediction interval. Additionally, `e0` is a shorthand wrapper for `flife.expectancy` with `age=0`. The m_x rates on which the life table is based can be the rates applying in a future forecast year t , in which case a period or cross-sectional life table is generated, or can be rates that are forecast to apply to a certain cohort, in which case a cohort life table is generated. All functions use the `cohort` argument to give cohort rather than period life tables and life expectancies.

For example, to generate the cross-sectional life table for males in 1980, we use:

```
> lifetable(usa, series="male", year=1980, type="period")
Period lifetable for USA : male

Year: 1980
      mx    qx    lx    dx    Lx    Tx    ex
0  0.0142 0.0140 1.0000 0.0140 0.9871 69.9883 69.9883
1  0.0011 0.0011 0.9860 0.0011 0.9854 69.0012 69.9835
2  0.0007 0.0007 0.9849 0.0007 0.9845 68.0157 69.0584
3  0.0006 0.0006 0.9842 0.0006 0.9839 67.0312 68.1089
4  0.0005 0.0005 0.9836 0.0004 0.9834 66.0473 67.1491
:
99 0.4066 0.3379 0.0051 0.0017 0.0043 0.0123 2.3891
100 0.4249 1.0000 0.0034 0.0034 0.0080 0.0080 2.3533
```

To generate the cohort life expectancy for males aged 65 by year in which aged 65 (using the coherent functional model obtained in Section 8.5), we use:

```
> usa.pr <- coherentfdm(smus1950.100, weight=TRUE, beta=0.05)
> usa.pr.f40 <- forecast(usa.pr, h=40)
> flife.expectancy(usa.pr.f40$male, age=65, type="cohort")
```

Point Forecast	
1976	14.45048
1977	14.54158
1978	14.63950
1979	14.75728
1980	14.88563
:	
2013	19.30161
2014	19.39335
2015	19.48484

To obtain prediction intervals for future life expectancies, we simulate the forecast log mortality rates as described in Hyndman & Booth (2008). Briefly, the simulated forecasts of log mortality rates are obtained by adding disturbances to the forecast basis function coefficients $k_{t,j}$ which are then multiplied by the fixed basis functions, $b_j(x)$ (assuming Equation 8.3). Then we calculate the life expectancy for each set of simulated log mortality rates. Prediction intervals are constructed from percentiles of the simulated life expectancies. This is all implemented in the **demography** package.

Using the coherent functional model obtained in Section 8.5, we can forecast period life expectancies (Figure 8.15) as follows:

```
> e0.fcast.m <- e0(usa.pr.f, PI=TRUE, series="male")
> e0.fcast.f <- e0(usa.pr.f, PI=TRUE, series="female")
> plot(e0.fcast.m, ylim=c(65,85), col="blue", fcol="blue", ylab="Years",
+ main="Product-Ratio method: coherent life expectancy forecasts")
> par(new=TRUE)
> plot(e0.fcast.f, ylim=c(65,85), col="red", fcol="red", main="")
> legend("topleft", lty=c(1,1), lwd=c(2,2), col=c("red", "blue"),
+ legend=c("female", "male"))
```

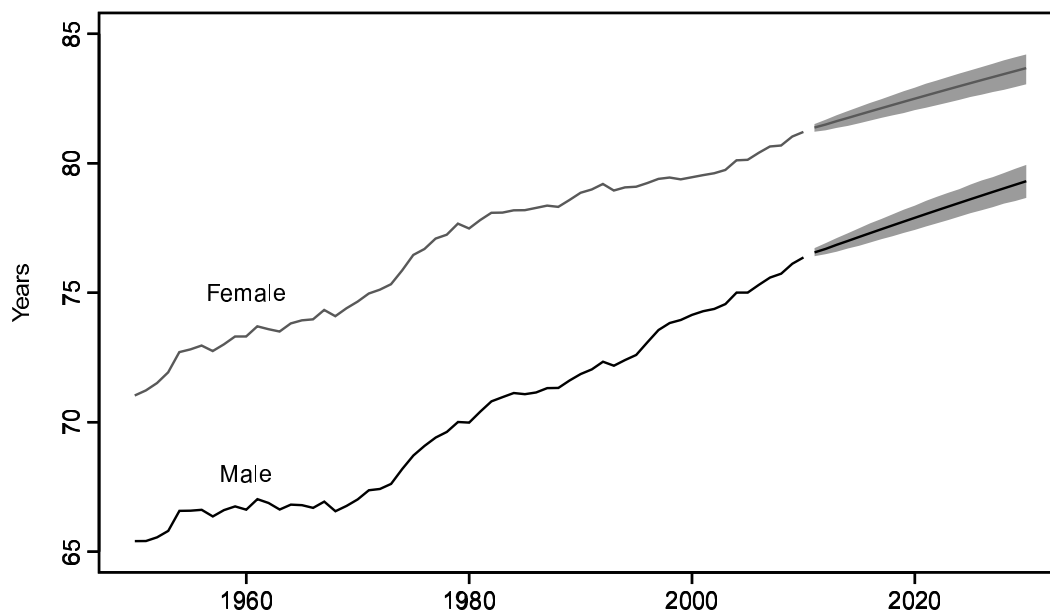
An alternative approach to life expectancy forecasting is direct modeling, rather than via mortality forecasts. This is the approach taken by Raftery et al. (2013) who use a Bayesian hierarchical model for life expectancy, and pool information across countries in order to improve estimates. Their model is implemented in the **bayesLife** package, available on CRAN.

8.7 Life Insurance Products

Connections between demographic functions, described in this chapter, and life contingencies (described in Chapter 7) can be made using the function **probs2lifetable()** in library **lifecontingencies**. The input used in this function is a vector of probabilities, either p_x or q_x .

To generate period q_x rates that can be input into the **lifecontingencies** package, we simply use the **lifetable** function for either observed (past) or forecast rates:

```
> lifetable1980 <- lifetable(smus1950.100, series="male", year=1980, type="period")
> qx1980 <- lifetable1980$qx
> lifetable2025 <- lifetable(usa.pr.f$male, series="male", year=2025, type="period")
> qx2025 <- lifetable2025$qx
```

**FIGURE 8.15**

Observed(1950–2010) and forecast(2011–2030) male and female period life expectancy using the product-ratio method with functional data models, United States.

If we want to use cohort probabilities, we will in most cases need to use both observed $m_{x,t}$ and forecast $\hat{m}_{x,t}$. Using the computations described above, and smoothed U.S. male rates at ages 0 to 100+ for 1950 to 2010, we obtain

```
> usa.pr <- coherentfdm(smus1950.100, weight=TRUE, beta=0.05)
> usa.pr.f150 <- forecast(usa.pr, h=150)
```

The matrix `smus1950.100$rate$male` contains past rates, $m_{x,t}$, and `usa.pr.f150$male$rate$male` contains forecast rates, $\hat{m}_{x,t}$. We define the combined matrix, `com.mx`:

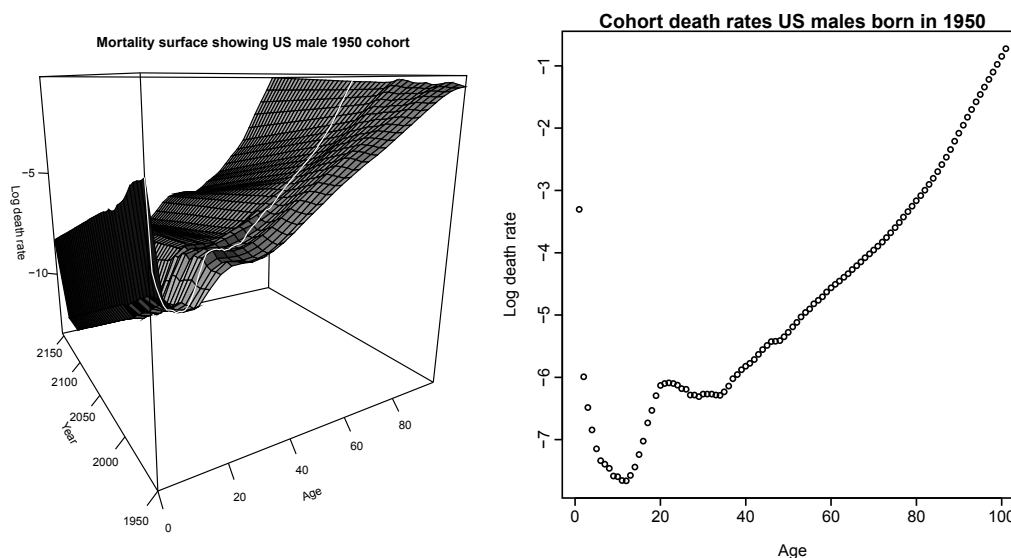
```
> com.mx <- cbind(smus1950.100$rate$male[1:nrow(usa.pr.f150$male$rate$male)],
+ usa.pr.f150$male$rate$male)
```

The cohort mortality rate for males born in 1950 is obtained using

```
> birthyear <- 1950
> mx <- com.mx[1:nrow(com.mx), (birthyear-1950+1):ncol(com.mx)]
> cohort.mx <- diag(mx)
> plot(log(cohort.mx), main="Cohort mortality rates, US males born in 1950",
+ ylab="Log death rate", xlab="Age")
```

Figure 8.16 shows the mortality rate surface as a function of x and t . The diagonal line is the cohort rate of mortality for males born in 1950.

```
> mat3d<-persp(seq(0,99,by=3),seq(1950,1950+209,by=5),
+ log(com.mx[seq(1,100,by=3),seq(1,210,by=5)]), theta=-30,
+ xlab="Age", ylab="Year", zlab="Log death rate", ticktype="detailed",
+ col=grey(.93), shade=TRUE)
> xyz3d <- trans3d(0:100,birthyear+0:100, log(cohort.mx), mat3d)
> lines(xyz3d, col="red", lwd=2)
```


**FIGURE 8.16**

Observed (1950–2010) and forecast (2011–2150) rates, showing cohort rates for U.S. males born in 1950.

Based on vector `cohort.mx`, we compute q_x using the equations given in Section 8.6. Note that Chapters 7 and 8 use slightly different assumptions for the construction of the life table, and therefore calculated life expectancy and other life table quantities will differ between the two for the same set of input q_x rates.

8.8 Exercises

- 8.1 Download the Human Mortality Database (HMD) mortality data for Denmark and plot male mortality rates at single ages 0 to 95+ for the 20th century.
- 8.2 Using data from 1950 for Danish females aged 0–100+, smooth the data by the Currie–Durban–Eilers and Hyndman–Ullah methods. Plot the two smoothed curves and the observed data for 1950 and 2000.
- 8.3 Download HMD data for Canada. Using data for the total population, compare forecast life expectancy for the next 20 years from the Lee–Carter and Lee–Miller methods.
- 8.4 Apply the Booth–Maindonald–Smith method to total mortality data for Canada. What is the fitting period? How does this forecast compare with the Lee–Miller forecast in terms of life expectancy after 20 years?
- 8.5 Using female data for Japan (from HMD), apply the Hyndman–Ullah method and plot the first three components. Plot forecast mortality rates for the next 20 years. How does the forecast differ from a forecast of the same data using the Lee–Carter method without adjustment?
- 8.6 Using male data for Japan, apply the Hyndman–Ullah method to forecast 20 years ahead, and plot male and female observed and forecast life expectancies on the same graph.

- 8.7 Apply the product-ratio method of coherent forecasting to data by sex for Japan. Plot past and future product and ratio functions. Add coherent male and female forecast life expectancies to the previous life expectancy graph.
- 8.8 Plot the sex difference over time in observed life expectancy, in independently forecast life expectancy and in coherently forecast life expectancy.